On „Cowgill's Law" in Greek

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1. INTRODUCTORY

1.1. According to J. Schindler, Gk. συρκ- (Hsch., *EM*) reflects an *o*-grade **tuorḱ-* beside zero-grade **turḱ-* in σάρξ „flesh", „avec *u* au lieu de *o* au voisinage d'une labiale" (1972b: 34, cf. 1972a: 97). A. Lubotsky has recently argued against this analysis, in the following terms (1994: 97): „it is unlikely that σύρκ- represents **tuorḱ-*, since this development has no parallels (**o* > υ in the neighborhood of labials always involves nasals, cf. νύξ, ὄνυξ, γυμνός, etc.). On the other hand, it seems conceivable that σύρξ reflects **twurk-* < **twr̥ḱ-*, cf. Lesb. πέσσυρες < **k*^{w}*etwurV-* < **k*^{w}*etwrV-* vs. Att. τέτταρες".

The **o* > υ process to which Lubotsky refers has now been codified under the name of „Cowgill's Law" (Sihler 1995: 42), having been described by W. Cowgill in the following simple formulation (1965: 156): „it seems clear that PIE **o* next to labials and labiovelars has become υ in a number of Greek words". Indeed, most of the examples discussed by Cowgill did involve nasals, leading Cowgill to rephrase the rule in terms of a change of „*no* to νυ" (op. cit. 157), on the basis of the forms ὄνυμα, νύξ, γυμνός, ὄνυξ, and various forms of στόρνυμι. Moreover, later treatments by Hamp (1970a, 1971a) have stated the **o* > υ process exclusively in terms of adjacent nasals: „[-round] > [+round] / [+nasal] ___ [+round]", i.e. „**o* > υ after nasal and before a rounded consonant" (1971a: 72). According to Hamp, moreover, the conditioning environment must be stated in „mirror-image" format, i.e. so as to include „/ [+round] ___ [+nasal]", thereby accounting for a case like πρυμνός „endmost" < **promno-*[1], showing that „the rounding lasted redundantly through the sonant segment".[2]

Nevertheless, Cowgill did not exclude non-nasal environments, as is clear from his discussion of φύλλον „leaf" (1965: 157n33): „Another Greek word with υ from **o* in labial environment is φύλλον, cognate with Latin *folium*, both from **bholyom*". On the basis of this example (and several similar ones, such as μύλη „mill" < **mol-eh*$_2$*-*, cf. Lat. *molīna* „id."), Sihler (loc. cit.) has most recently generalized the rule as follows: „**o* from any source > υ in Greek between a labial and a resonant".

1.2. The last two examples cited evince the connection between the process in question (however it is to be formulated) and a long-standing problem surrounding a number of words with „irregular υ" in Greek[3]: traditionally, many such forms are assumed to reflect not an *o*-grade, but rather an „alternative" zero grade treatment,[4] or an irregular zero-grade treatment with „reduced vowel" or „alternative treatment of the zero-grade prop vowel",[5] whence handbook notations of the sort **bh^{u}l̥io-* or **bh^{o}l̥io-* for φύλλον, **m^{u}l-eh$_2$-* or **m^{o}l-eh$_2$-* for μύλη, etc. (and compare Lubotsky's account of Lesb. πέσσυρες, 1.1.).[6] It should be noted at the outset that such an approach does not constitute an explanation: it merely characterizes an intractable irregularity, namely an „alternative" zero-grade treatment that surfaces under unknown circumstances – in principle, a construct one would like to be able to do without.

1.3. Another class of Greek „irregular υ" forms related to the **o* > υ problem concerns putative *e*-grades: thus κύκλος „circle" and γυνή „woman", both beginning with PIE labiovelars, are traditionally said to show a rounding process *K^{u}e* > *K^{u}o* (Schwyzer 1939: 296), and then to undergo a subsequent **o* > υ process, although only γυνή, of course, involves an adjacent nasal.[7]

The purpose of this paper, then, is to reconsider the evidence bearing on the formulation of one or more **o* > υ rules in Greek, together with their implications for a series of related problems concerning Greek „irregular υ" more generally.

2. On Cowgill's Law „sensu stricto" (nasal environments)

2.1. *On exceptions.* Consider first Cowgill's Law in the strict version envisioned by Lubotsky, i.e. environments with nasal, especially environments with labial stop and nasal. As Dunkel has noted (1982b: 58n16), it is not difficult to find numerous counterexamples to the alleged **o* > υ process, most of which can be explained away on the basis of trivial analogies or easy restorations: one can readily account in this way for *o*-vocalism (not *u*-vocalism) in e.g. thematic acc. sg. and 1 sg. impf./aor. forms in -μον /-βον /-πον (ἵππον, not †ἵππυν; ἔλαβον, not †ἔλαβυν; etc.) and in various parts of thematic paradigms (ἔπομαι, not †ἔπυμαι, etc.), in the paradigms of μον-stems and (ϝ)ον-stems, and in forms involving (at least semi-)productive *e/o* ablaut types (πομπή „an escort" : πέμπω „send", μόνιμος „steadfast" : μένω „remain", etc.). Still other potential counterexamples could be eliminated as belonging to obscure alternation types (e.g. μονθυλεύω „stuff [food]" beside ὀνθυλεύω „id."), or as evident borrowings that could have escaped the operation of the rule (e.g. βόνασ(σ)ος / μόναπος „bison, aurochs"). Certain more isolated material, however, is at least disquieting. Thus e.g. φονή, φόνος „slaughter" and the onomastic formant -φοντης „-slayer" (cf. Myc. *-qo-no, -qo-ta,* Leukart 1994: 51ff.) are not directly derived from the reduplicated aorist πεφνεῖν (the only other related form with stem in [-]φ-), and would appear to be relatively old formations that should have undergone an early **o* > υ process, as should *a fortiori* φοινός „(blood-)red" (**φονι̯ο-*), an old word which may well have only a folk-etymological relationship to φονή etc. (on which see recently Watkins 1995: 373).

Still more troubling is the isolated (and inherited) word πόντος „sea", the only item acknowledged by Hamp (1971a: 72n5) as a potential counter-example to the process in question. Hamp claims that „πόντος was paradigmatically preserved or restored by πάτος", but this reasoning is not clear. Whatever the precise history of this word in Greek and earlier,[8] one clearly has to reckon with a very old paradigm split that resulted in an *o*-grade thematic **pont(-H)-o-* of sufficient antiquity to yield †πύντος (according to the version of the rule stated above), and not likely to be susceptible to influence from the zero grade that resulted in πάτος. If, then, πόντος constitutes a serious counterexample, the simplest adjustment would be to restrict the **o* > υ process in labial stop plus nasal contexts so as to apply to *labiovelars only,* as in the clear cases νύξ „night" (with sequence **nok^{u}t-*) and γυμνός „naked" (with ambiguous preform, given well-known problems with this word involving tabu deformation at both the IE and einzelsprachlich levels, as well as other factors,[9] but with **g^{u}on-, *g^{u}om-, *nog^{u}t-* or **mog^{u}t-* assured as the pre-Greek input).

2.2. *Nasal + nasal type (i.e. /-nom-/: ὄνυμα „name", 1 pl. στόρνυμεν „we spread" etc.).* The fact that various Greek words in νομ- show a somewhat obscure semantic relationship to their base verb νέμω „distribute" is a general problem,[10] which need not prejudice unduly the evaluation of such forms. Since their formal derivation is in general clear, such forms, even though semantically to some extent „isolated", do not constitute serious counterexamples to an **o* > υ process in this context.

As for the υ-vocalism of the more clearly isolated form ὄνυμα (beside which ὄνομα is clearly due to a later assimilation), this is sometimes ascribed to analogical influence from compounds that could in principle show an ω … ο > ω … υ dissimilation, as in Hom. νώνυμ(ν)ος „nameless" (thus e.g.

Lejeune 1972: § 192 n. 3, comparing Argolic οροφᾶ / υπ-ωρυφιᾶ). But even if this case is equivocal, there appears to be no principled way of accounting for the development of nasal presents like στόρνῡμι (cf. στρωτός < *$str̥h_3$-tó-, aor. στορε- < [metathesized] *stero- < *$sterh_3$-, and Ved. *str̥nā́ti*) without recourse to the development /-nom-/ (< *-*n*-h_3-*m*-, with vocalized laryngeal) > -νυμ-, as convincingly argued by Cowgill (1965: 154ff.).[11]

It will not do, however, to assume, with Hamp (cf. 1.1.), that the *o > υ rule would operate on /mon-/ just as readily as on /-nom-/, given the isolated word μόνος (Ion. μοῦνος) „alone" (< *$monu̯o$-, with labial glide after the second nasal!).

2.3. *πρυμνός type* (cf. 1.1.). At first sight, a possible counterexample would seem to be μορφή „form, shape", which is almost completely isolated; nevertheless, the gloss ἀμερφές: αἰσχρόν Hsch. could point indirectly to a *μέρφω (unattested as such), which would serve to weaken the value of μορφή as a counterexample (cf. πομπή : πέμπω etc., 2.1.; with phonetic shape as in μορφή : *μέρφω, likewise πόρνος / πόρνη „prostitute" : πέρνημι „sell", and similarly φρον- [e.g. φρονέω „think"] : φρένες „seat of understanding"). More troubling, however, is the entirely isolated *o*-grade μορμώ „monster, scarecrow, nightmare", which belongs with Lat. *formīdō* „fear; scarecrow" (with *u*-vocalism attested only in Hsch. μύρμος· φόβος, beside Hsch. μόρμη· χαλεπή, ἐκπληκτική).[12] It is important, then, to consider alternative explanations for the *u*-vocalism of πρυμνός. Indeed, an acceptable one can be found: according to G. Dunkel's conception of ablaut in particles (1982/1983 and later publications, esp. 1992: 161ff.), the *u*-vocalism simply continues an inherited variant *pru* (beside *pro*, cf. διαπρύσιος „penetrating" beside διαπρό „thoroughly"; thus explicitly Dunkel 1982a: 54n9, 1992: 161n32), as in *$k^{u̯}o$- / *$k(^{u̯})u$-, *apo / *apu* etc.[13] Sihler (1995: 42) adduces μύρμηξ „ant" (: Lat. *formīca* etc.) as another example of the πρυμνός type; but *u*-vocalism in a form of this sort, which shows widespread variation or deformation throughout Indo-European,[14] does not support this version of the rule very strongly, especially since *o*-vocalism is attested with some certainty even for Greek (Hsch. βόρμαξ, ὄρμικας [i.e. *ϝορμ-]).[15]

At best, then, the testimony of πρυμνός and similar forms is essentially equivocal; thus the only good evidence for „Cowgill's Law" so far remains the type with nasal and labiovelar (νύξ, γυμνός) and the type with /-nom-/ (ὄνυμα, στορνῦμ-).[16]

2.4. *ὄνυξ „nail"*. The form ὄνυξ, although superficially similar to νύξ and widely assumed to have its υ via Cowgill's Law, poses a rarely-acknowledged difficulty.[17] For those who, like Cowgill himself (followed by Sihler), explicitly reconstruct the form as *$h_3nog^{u̯}h$- with labiovelar, there is no problem whatsoever: ὄνυξ would in this case behave exactly like νύξ and γυμνός. A problem arises, however, for those who reconstruct the form *without* labiovelar, and who nevertheless refer to Cowgill for the *u*-vocalism: thus e.g. Rix (*h_3nogh-*u*-/*$h_3n̥gh$-*u*-, *h_3nogh-*lo*- etc., 1970: 108n78), and Beekes (*h_3nogh- [1969: 47] with „o > υ in the vicinity of labial sounds", citing νύξ, πρυμνός, and Cowgill), both in influential treatments.[18] But with a reconstructed root shape *h_3nogh-, there is no „labial sound" (apart from *h_3-) that is available to trigger the o > υ process; and despite Cowgill's informal characterization in terms of a „*no* to νυ" process (cf. 1.1.), an unconditioned change /no/ > /nu/ is patently out of the question, in view of counterexamples involving isolated and old-looking items of the type νόθος „bastard", (Ion.) νόος „mind", νόσος „sickness", νόσφι „apart from", Νότος „south(west) wind". Since the IE word for „nail" happens to show considerable formal variation, the simplest approach would be to assume, with Cowgill, that at least for Greek (though clearly not for all of the rest of IE: cf. Ved. *nakhā-*, Germ. *Nagel* etc.), the preform to which Cowgill's Law applied had a labiovelar.[19]

In order to account for ὄνυξ *without* labiovelar, one could begin with the *u*-stem *$h_3$nógh-*u*- reconstructable on the basis of material outside Greek (OPr. *nagutis* etc.), and suppose that a generalized acrostatic inflection (cf. the type γόνυ, γουνός / γούνατος „knee") resulted in a paradigm *$h_3$nógh-*u*-*s*, *$h_3$nógh-*u*-*os*, *$h_3$nógh-*u*-*i* etc.; the sequence *-*nogh-u̯*- might then have been treated like *-*nog$^{u̯}$h*- (cf. *$ǵ^{u̯}h$* treated like *$g^{u̯}h$* in θήρ, Aeol. φήρ „wild animal"), and the root-noun inflection in Greek would be taken as secondary.[20] What is slightly problematic for this account is the failure of „Saussure's [and Hirt's] Law"[21] to delete the initial laryngeal in this consistent *o*-grade context. Alternatively, one could assume that the generalized *o*-grade was formed after the laryngeal was vocalized, the laryngeal itself having been generalized from the weak stem of an earlier *$h_3$nógh-u- / *h_3n̥gh-(e)u-*. At bottom, however, there is no actual evidence for *u*-stem inflection in Greek. It would be simpler to take the Greek root noun inflection at face value, as an isolated morphological archaism (Schindler 1972a: 91). As Alan Nussbaum points out to me, the failure of Saussure's Law to affect *h_3- could then be ascribed to influence from the weak stem of the presumed root noun with *o/e* ablaut underlying ὄνυξ, prior

to generalization of the *o*-grade (cf. **pod- / *ped-* „foot" [Schindler 1972b: 33f.] vs. Gk. πούς / ποδός). In this event, however, the reconstruction with labiovelar becomes necessary, since a labial conditioning would again be otherwise lacking.

2.5. *κύνα (acc. sg.) „dog"*. Sihler (1995: 294), operating with the highly general version of Cowgill's Law cited in 1.1. (ad fin.), makes the interesting suggestion that acc. sg. **ḱu̯ón-m̥* could have resulted more or less directly in κύνα (with κ- paradigmatically restored for *πύνα, assuming **ḱu̯-* > π[π]-), so that this form might actually preserve the inherited *o*-grade of Ved. (acc.) *śvā́nam*, and not the zero grade of the weak stem (e.g. gen. κυνός, cf. Ved. *śúnas*). For Sihler, the conditioning labial would be the labial glide in the sequence /-won-/; but it will be shown below (3.1.) that a sequence /-won-/ is of itself incapable of triggering a development to /-wun-/. What is suggestive at this point is the phonetic similarity between the sequence **ḱu̯ón-* and the other Cowgill's Law cases involving labiovelar (**nok\u̯-* „night", [probably] **nog\u̯-* or **g\u̯on-* „naked", **h₃nog\u̯h-* „nail") or (as a possibility for „nail") velar plus labial glide (**h₃nogh-u̯-*). Thus (apart from the problem of the restored initial *k-*) it might be possible to consider **ḱu̯ón-* (> κυν-) as reflecting essentially the same phonetic environment as in the bona fide Cowgill's Law cases **nok\u̯-* (> νυκ-) etc.; nevertheless, a somewhat different conception will be advanced below (5.1.).

2.6. *γυνή „woman" and κύκλος „circle, wheel"*. As already noted (1.2.), γυνή and κύκλος are traditionally said to continue *e*-grades, which then undergo Cowgill's Law via an „intermediate" *o*-grade (i.e., **K\u̯e* > **K\u̯o*, then **o* > υ). If indeed κύκλος is based on an old *e*-grade (cf. Ved. *cakrá-* etc.), such a development would have a high degree of phonetic plausiblility in this case, since the **e* would be flanked by two labiovelars, and might well have undergone at least one process of rounding.[22] In contrast to **k\u̯ek\u̯-* > **k\u̯ok\u̯-*, a process **g\u̯en-* > **g\u̯on-* (as claimed for γυνή) has no strong claim to phonetic naturalness, and is supported by no comparable behavior in Greek: cf. **g\u̯hen-i̯e/o-* > θείνω (not †θοίνω or †θῦνω), without **e* > **o*, as well as the general failure of **e* > **o* even for sequences of labial + labiovelar (e.g. [ϝ]ἔπος „word, speech" < **u̯ek\u̯-*) and labial + labial nasal (e.g. [ϝ]ἐμέω „vomit" < **u̯emh₁-*). Even if the υ of κύκλος has a different source, i.e. through labial coloring of the reduced vowel (otherwise normally resulting in Gk. ι, cf. πίτνημι „spread out" < **pₑt-néh₂-*) in the „schwa secundum" context produced by the zero-grade collective **kₑ\u̯kₑ\u̯léh₂* (> κύκλα),[23] this would not affect the point at issue: we would again have a

phonetic environment equally (and perhaps even more) conducive to rounding effects on the vowel in question, unlike the presumed **e* in the preform of γυνή.

There are only three other alleged cases of a general process **K\u̯e* > **K\u̯o* (Schwyzer 1939: 296), of which two are not remotely convincing, and the third is problematic:

(i) σκύζα „lust, passion", with labiovelar assumed on the basis of Hsch. σπάζει· σκυζᾷ. Ἀχαιοί – but σπάζει belongs rather with σπάω „draw, pluck, tear";

(ii) σκύλαξ „puppy", comparing ON *skvaldra* „shout, chatter", mod. Norw. also „yelp (etc.)" – but *skvaldra* belongs with *skvala* „rush (of streams etc.)" > „chatter", thereby undermining the etymology of σκύλαξ with labiovelar;[24]

(iii) βυθός, βυσσός „the deep/depths (of the sea)" (with β- for expected γ-, cf. γυθίσσων· διορύσσων Hsch.), cf. βαθύς „deep" and allegedly also βῆσσα „glen, vale" – but the root vocalisms of these forms are not directly comparable to each other (note βένθος „depth (of the sea)" beside βαθύς), and moreover, it is difficult to separate the *u*-vocalism of βυθός/βυσσός from the (inherited) *u*-vocalism of πυθμήν „bottom" (also πύνδαξ „bottom [of a vessel]"; cf. Hamp 1990: 448f.), which could well have exerted some influence on whatever form underlies βυθός / βυσσός.

There being, then, virtually no evidence to support a development **g\u̯en-* > **g\u̯on-*,[25] we may reconsider the morphology of γυνή, to see whether *e*-grade is in fact the necessary starting-point. In general, the form of the word for „woman" directly underlying γυνή is assumed to continue a paradigm with *e*-grade / zero-grade ablaut (nom. **g\u̯en-éh₂*, gen. **g\u̯n̥(n)-éh₂-s* etc.) comparable to the alternation found in Arm. *kin/kan-*, with Boeot. βανᾶ reflecting the zero-grade allomorph.[26] There are thus two traditional sources for the υ of γυνή, namely the *e*-grade (which, though morphologically unproblematic, requires unsatisfactory ad hoc phonetic assumptions, as just discussed), or alternatively via the zero grade. For the latter, Rix (1976: 149) suggests that an original ablauting paradigm such as **denā/*gunās* was remodeled to **gunā/*gunās*, with the weak stem generalized.[27] Although neither Rix nor de Lamberterie (1991) provides any phonetic explanation for the development **g\u̯n̥(n)-* > /gun-/, one could assume that the (presumably schwa-like) zero-grade „prop vowel" arising from the syllabic nasal was rounded to /o/ (thus **g\u̯n̥(n)-* > **g\u̯ᵊn(n)-* > **g\u̯ᵘᵒn-*); the resulting sequence would then have corresponded to the „strict labiovelar" context of Cowgill's

Law as discussed above, i.e. *$g^{\mu o}n$- ~ *$g^{\mu}on$- or *nog^{μ}- „naked", *nok^{μ}- „night", etc. But this is only a minimal improvement on the *e*-grade scenario: one is still left with a phonetically unmotivated „intermediate" *o*-grade, with no independent support; indeed, it is particularly troubling that Boeot. βανᾱ shows (at least on the simplest interpretation of the form) the expected treatment of the syllabic nasal.[28]

As an alternative, one may suggest, with due reserve, that γυνή simply continues an *o*-grade, which would then be subject to the „strict" Cowgill's Law context seen in νύξ and γυμνός. In the abstract, such an *o*-grade might well be viewed as „scarcely credible" (thus Hamp 1970a: 486), except that one could appeal to Anatolian (Luv. *wāna-*, Lyd. *kāna-*) for independent support.[29] And from an Indo-European point of view, the resulting *g^{μ}*ón-h*$_2$/*g^{μ}*n-éh*$_2$*-s* (beside *g^{μ}*én-h*$_2$/*g^{μ}*n-éh*$_2$*-s*) might not require undue complications (cf. patterns of the sort **gén-u/*gón-u* „knee"). One is reluctant, nonetheless, to posit such complex allomorphy for Greek, where both *e*-grade forms (de Lamberterie 1991) and zero grade forms (at least Boeot. βανᾱ[30]) must also be assumed.

2.7. *σύν/ξύν „with"*. G. Dunkel (1982b) asserts that, despite the evidence of Myc. *ku-su-* and the text of Homer, σύν is older than ξύν, and that σύν derives from **sóm* (= Ved. *sám*, cf. **sem-* „one, united"). This derivation requires not only the irregular Greek treatment **s-* > σ-, but also an extended version of Cowgill's Law, which is now allowed to operate on sequences with nasal and /-s-/ („*o* > *u* / *nasal* and *liquid, labial, labiovelar, *s*", in Dunkel's formulation). The only other example Dunkel cites is the titular term αἰσυμνήτης „judge, magistrate" (etc.), in Saussure's interpretation **aiso-mnā-* (see Frisk s.v. αἰσυμνάω with Nachtr.; on the meaning, see recently Ford 1992: 116, Campanile 1993: 28); but the form is etymologically obscure, and could well be borrowed.[31] Dunkel provides, moreover, no cogent phonetic justification for this reformulation, appealing only to restricted (and not precisely comparable) dialectal phenomena of the sort Arc. ὐν- (for ὀν-), Lesb. ὐσδ- (for ὀσδ-), etc.[32] There are, finally, three sorts of potentially troubling counterexamples, none of which Dunkel mentions:

(i) for forms with **s-* > /h-/ or Ø, **som-* never develops as †*hum-*, as in other *o*-grade material from IE **sem-* (ὁμός „same, common", ὁμοῖος „similar", ὁμαλός „like"), as well as in ὅμαδος „din (of battle etc.), (noisy) throng" (: Ved. *samád-* „battle"),[33] and ὁμφή „divine voice, melody" (: Eng. *song* etc.);

(ii) surface /som-/ is an acceptable sequence for Greek, cf. σομφός [Hippoc.+] „spongy, porous" (although the testimony of this form is subject to certain qualifications, as discussed in another connection below, 4.8.);

(iii) there is no evidence for a „mirror-image" version of Dunkel's rule (i.e. **Nos-* > †*Nus-*), as shown by forms like μόσχος „calf" (: Arm. *mozi* „bullock" or Li. *mãzgas* „bud", cf. Clackson 1994: 152ff.) and νόσος / Ion. νοῦσος „sickness, disease" (< **nosu̯o-* or **nosso-*).

Since, then, Gk. σύν has other explanations that do not require so many difficult assumptions and that fit the philological facts without special pleading,[34] I conclude that there is insufficient evidence for the expanded version of Cowgill's law operating on sequences with nasal and /-s-/.

3. MÝΛΗ, ΦΎΛΛΟΝ ETC. AND COWGILL'S LAW „SENSU LATO" (**O* > Y BETWEEN LABIAL AND RESONANT)

3.1. *On exceptions.* Consider now the still more general claim of Dunkel, Sihler and ultimately Cowgill that forms like μύλη and φύλλον show an **o* > υ process conditioned by flanking labial and resonant, including non-nasal resonants. Such a formulation immediately faces a host of counterexamples. Again, some are apparent only, or eliminable with recourse to easy or relatively easy analogies,[35] or negligible as more or less clear borrowings;[36] and still others could be described as disquieting, but perhaps ultimately not probative.[37] There remains, yet, a substantial residue of cases that are difficult or impossible to explain away. The material for **u̯or-* and **u̯ol-* is particularly abundant, as shown by the following isolated forms (all Hom.+):

(i.1) Pr.-Gk. **u̯or-* > (ϝ)ορ-, not †(ϝ)υρ-:
ὀρθός „upright" (**u̯or[H]dhu̯ó-*,[38] cf. zero grade in Ved. *ūrdhvá-*);
ὄρτυξ „quail" (Hsch. γόρτ- = ϝόρτ-, cf. Ved. *vártikā-* „id.");
οὖρος „guardian" and other forms related to ὁράω[39];
(i.2) Pr.-Gk. **u̯ol-* > (ϝ)ολ-, not †(ϝ)υλ-:
ὦλκα „furrow (acc.)" (**h$_2$u̯olk-*, cf. αὖλαξ, Schindler 1972a: 77);
οὐλή „scar" (**u̯ol[h$_2$]-néh$_2$* [laryngeal loss as in ὀρθός], cf. Lat. *vulnus*);
οὖλος „woolly" (**[H]u̯ól[H]-no-* [laryngeal loss as in ὀρθός], cf. λῆνος „wool" etc.);

o-grade forms related to **u̯el(H)-* „roll, turn",[40] e.g. ὅλμος „rolling-stone, mortar" (**u̯ól[H]-mo-*), οὐλαμός „throng (of warriors)" (ϝολ- with metrical lengthening, cf. Hsch. γολ- = ϝολ-).

Nor is there any lack of material for other sequences, e.g.:

(ii.1) Pr.-Gk. **mor-* > μορ-, not †μυρ-:
μόρον „mulberry"[41] (cf. Arm. *mor*, Lat. *mōrum*);
(ii.2) Pr.-Gk. **mol-* > μολ-, not †μυλ-:
μολύων „sully, defile", river name Μολόεις (cf. Ved. *mála-* „filth")[42];
μολγός „leather bag" (cf. **mólko-* „id." in Gmc., IEW 747);
(iii.1) Pr.-Gk. **por-* > πορ-, not †πυρ-:
πόρις / πόρτις „calf" (: Arm. *ort'* etc.);
(iii.2) Pr.-Gk. **pol-* > πολ-, not †πυλ-:
πόλτος „porridge" (cf. Lat. *puls* „id.")
(iv) Pr.-Gk. **hroph-* > ῥοφ-, not †ῥυφ- (except dialectally):
ῥοφέω / ῥοφάνω „swallow" (: Lat. *sorbeō* „id." etc.).[43]

Similarly, there are many other isolated forms with less clear Indo-European pedigrees, but with every chance of being relatively old within Greek; such forms (or at least some of them) should have been likely candidates for the application of such a rule, but have failed to undergo it, e.g. Hom. μόρφνος (epithet of eagle), Hom. (ϝ)ὄρχος „row of vines / trees", Hom. μολοβρός (perh. „gluttonous"), Hom. λόφος „crest", Hom. φολκός (B 217, Thersites; e.g. „bandy-legged"), Hom. φορύνομαι „become mixed / sullied", Hom. λοβός „lobe"; trag.+ μόλις adv. „scarcely", φολίς „scales (of a reptile), spots / patches (on animal hide)" (cf. φόλυες κύνες· οἱ πυρροὶ ὄντες μέλανα στόματα [στίγματα] εἶχον, Hsch.), among a number of other such forms that could be adduced.

3.2. *μύλη and φύλλον.* In view of the above listings (not exhaustive, but surely sufficient to establish the point), it becomes impossible to grant that μύλη owes its -υ- to a regular application of this generalized version of Cowgill's Law. Indeed, at least one alternative account is available. E. Hamp has suggested (1983: 12, 1991: 87), in a not implausible way,[44] that the *u*-vocalism of this form, as in some others with odd *u*-vocalism, is traceable to a „Prehellenic" origin.[45] (Another way of accounting for the vocalism of μύλη will be suggested in 3.3.1. below.)

Similarly, the weight of the counterexamples marshaled in 3.1. militates against a simple **o* > υ account for φύλλον. While Lat. *folium* (very likely to continue the same formation as φύλλον) is ambiguous between zero grade and *o*-grade, the latter is surely the more attractive working hypothesis: given our interest in rejecting the traditional conception of a *u*-colored Schwundstufe in φύλλον and similar forms (note also the root accent of φύλλον, favoring a full grade), all that is needed is a reasonable means of accounting

for φύλλον as an *o*-grade.[46] Under the assumption of an original *o*-grade, and in view of the broader phonetic environment shown by this form, I suggest that the υ of φύλλον results not from a Cowgill-like **o* > υ process involving a neighboring labial (which, as shown above, is attested with certainty only for /noK$^{\underline{u}}$/ ~ /K$^{\underline{u}}$on/ and /nom/), but from a raising of **o* > υ consequent on the palatalization (with gemination) of the /-l'/ in the position before yod, i.e. (in informal notation) **-oli̯-* > (**-ol'l'-* >) **-ull-*.[47] If this were a phonologically regular process, we would have the following two expectations (under ideal circumstances):

(i) we would expect to find other cases of Gk. -υλλ- < **-oli̯-* (although naturally not all Greek forms with surface -υλλ- will have this source);

(ii) we should be able to demonstrate that surface -ολλ- does not derive from **-oli̯-* (which might otherwise be expected, given standard assumptions about the development of /l/ plus yod sequences in Greek). The remaining two subsections of 3. address these two points in turn.[48]

3.3 *-υλλ-* < **-oli̯-*. The following items in -υλλ- seem to me to have a reasonable chance of showing the development just suggested for φύλλον; all have hitherto been considered phonologically problematic (precisely because of suspected connections with related material showing ordinary full grade or zero grade), but can in this way simply reflect *o*-grades.

3.3.1. The substandard and isolated verb μύλλω 'βινέω' (Theoc.; cf. Sicil. μυλλός „Kuchen in der Form der pudenda muliebra") is generally assumed to belong with the root **melh$_2$-* „grind", despite the difficult *u*-vocalism. According to the above hypothesis, it could directly continue an *o*-grade, which would be quite in keeping with the profile of this prototypical acrostatic *o/e*-verb, according to the well-known conception of Jasanoff (1979 and later publications: cf. Go. *malan*, OCS *meljǫ* etc.).[49] Thus μύλλω would reflect a **mol[h$_2$]-i̯e/o-* (with regular laryngeal loss), beside the *e*-grade in Myc. *me-re-ti-ri-ja* „corn-grinders", *me-re-u-ro* „flour", and not an obscure „schwund- oder reduktionsstufige Variante" (Frisk).[50]

As a consequence of the above, it seems possible that an original **μόλη* (cf. Lat. *molīna*) was remodeled to μύλη [Hom.+, cf. Hom. μυλοειδής, μύλαξ, Hes. μυλιάω] after μύλλω, at a (relatively early) time when this still retained its technical meaning. The general replacement of **melh$_2$-/*molh$_2$-* with ἀλέω (apart from survivals like *me-re-ti-ri-ja / me-re-u-ro* and μύλη), and the consequent development of a specialized meaning for μύλλω, could

in turn be associated with the formally anomalous allomorphy /mela-/, /mele- / ~ /mul-/ for this *o/e*-verb produced by purely phonological means.

3.3.2. In a similar way, σκύλλω [Aesch.+] „flay, rend, snatch violently etc." (and probably the monster Σκύλλη, cf. also σκάλλω [Hdt.+] „hoe, harrow") need not show „une coloration particulière du vocalisme zéro" (Chantraine, cf. the „Reduktionsvokal" of Frisk and others); instead, a plain *o*-grade **skol[H]-i̯e/o-* could be juxtaposed more or less directly with Hitt. *iškalla-* „tear" (beside Li. *skélti* „split" and other forms), for which Jasanoff (loc. cit.) likewise reconstructs an *o/e*-verb.[51]

3.3.3. κυλλός „crooked, crippled" (cf. Hom. κυλλοποδίων „lame" [Hephaestus]) appears to belong with other similar terms referring to physical deformities, all with normal *e*- or *o*-grade – thus e.g. Frisk s.v. κυλλός: „Wohl mit κελλόν· στρεβλόν, πλάγιον H. […] irgendwie zusammenhängend", cf. also Hsch. κελλάς· μονόφθαλμος (as well as cognates elsewhere, e.g. OIr. *coll*, Ved. *kāṇá-* „one-eyed"). κυλλός, then, could continue an *o*-grade end-stressed adjective **kol-i̯ó-* (beside **kel-i̯/nó-*, **kol-nó-*).[52]

M. Meier-Brügger (1990; 1992: II.24) has recently offered a different etymological account. Taking κυλλός to mean essentially „twisted", he proposes connection with **kʷel-* „turn"; thus κυλλός would reflect a **kʷl̥-nó-*. This calls for the following observations:

(i) Meier-Brügger's approach has the undesirable consequence of separating κυλλός from κελλόν etc.;

(ii) there is a growing consensus that **kʷel-* „turn" is actually set **kʷelH-*,[53] probably including Greek (e.g. πλήμνη „nave of a wheel" [Hom.+]),[54] so that an anit **kʷl̥-nó-* cannot be set up without further justification;

(iii) a zero-grade **kʷl̥-nó-* cannot in any event explain the *u*-vocalism of κυλλός: despite Meier-Brügger,[55] the phonological context does not admit an appeal to „schwa secundum" (as opposed to the canonic context seen in forms like πίτνημι, cf. Mayrhofer 1986: 175ff. and Peters 1980: 98f., after Schindler 1977: 31);

(iv) an account via **kʷel-* (in some ways semantically attractive, as Meier-Brügger shows) could in any event be salvaged by appealing to the phonological development under consideration, according to which an *o*-grade **kʷol[H]-i̯ó-* should yield κυλλός regularly.

3.3.4. Among several other more problematic cases, the following two may be mentioned briefly:

(i) The gloss θύλλα· κλάδους, ἢ φύλλα· -ἢ ἑορτὴ Ἀφροδίτης (Hsch.)[56] seems clearly to belong with the family represented by such forms as θαλλός „young shoot, branch", θάλλω „sprout, thrive" (cf. Alb. *dal* „id."), θαλερός „sturdy, luxuriant" (cf. Arm. *dalar* „green, fresh").[57] Unfortunately, the root shape to be reconstructed for these forms is notoriously unclear, both in terms of root vowel (IE **a* vs. „Leiden-style" **h₂*), as well as the presence or absence of a final **-h₁*.[58] To the extent that an *o*-grade could be justified in both phonological and morphological terms (which is not readily apparent), the form could in theory reflect a (Pr.-Gk.) **tholi̯a* or even **thōli̯a*,[59] granting **-ōli̯-* > -υλλ-. Morphologically (and semantically), one could consider the form in terms of a collective built to an acrostatic *i*-stem (thus roughly **dhol(h₁)i-h₂* or **dhoh₂l(h₁)i-h₂* „sprouts, branchwork"), although a devī-formation based on a root noun would produce the same formal result.[60] Given, then, the various indeterminacies involved, θύλλα cannot strongly support the process in question.

(ii) Beside βδέω „fart", Greek at some point created a full grade βδελ- (e.g. βδελυρός [Aristoph.+] „disgusting"), with parallel *o*-grade in βδόλος [Com. Adesp.] „stench". The clearly related verb βδύλλω [Aristoph.+] „be in deadly fear of" (Chantraine: „lâcher un vent de peur") is normally assumed to be formed „avec le suffixe familier -ύλλω" (Chantraine), but could directly reflect a relatively old *o*-grade **bdoli̯ō* related to βδόλος. In view of the expressive possibilities available, a form like βδύλλω cannot be used to support **-oli̯-* > -υλλ- (which in any case seems adequately justified by φύλλον, μύλλω, σκύλλω and κυλλός).

3.4. *Surface -ολλ-*. The relatively rare words with surface /-oll-/ can be categorized as follows[61]:

3.4.1. *Words of obscure origin.* Most cases have the appearance of being loans with unidentified source:

(i) σκερβόλλω „scold, abuse" (Aristoph.) (cf. σκέρβολος „scolding, abusing" [Callim.]) and related glosses (κερβόλλουσα, σκέραφος, κέραφος); second member perhaps related to βάλλω, but in any case, „expressive Wörter unklarer Bildung und dunklen Ursprungs" (Frisk);

(ii) a series of words with initial sequence κολλ- (cf. [iii] and 3.4.4.[i] below), and with partly overlapping semantics; some items have traditionally been compared with κόλλα (3.4.4.[i]), but all are in fact quite obscure:

(ii.1) κόλλοψ „screw, peg (esp. of a lyre)" – „Fachausdruck unbekannter Herkunft" (Frisk);

(ii.2) κόλλαβος 'κόλλοψ', also „a kind of bread/cake"; cf. also κολλαβίζω „play the game of κόλλαβος"–„Volkstümliches Wort auf -βος [...] ohne Etymologie" (Frisk);

(ii.3) κόλλιξ „large round bread", also „tablet" (in medical writings)–„wie bei κόλλαβος [...] müssen wir aus Unkenntnis der Tatsachen auf eine Erklärung verzichten" (Frisk; see Nachtr. 133 s.v. for possible Iranian source forms);

(ii.4) κολλῦρα (also -ουρ-, cf. [iii] below) „cake; tablet" (cf. κόλλιξ), and diminutives (e.g. κολλύριον) generally „eye salve" (i.e. *„small tablet") – „Bildung wie ἄγκῡρα, λέπτῡρον u. a. [...], sonst dunkel" (Frisk);

(iii) κόλλουρος (name of an unknown fish; see Frisk s.v. for Strömberg's analysis, involving expressive gemination).

3.4.2. *Loans.* Source forms have been identified for the following two probable loans (although not without difficulties in each case):

(i) κόλλυβος (Eup.+) „small change", later also „gold weight" and „exchange rate", cf. κολλυβιστής (Men.+) „money changer" etc.; perh. cf. Hebr. *hālap* „exchange",[62]

(ii) ἀβόλλης (attested only in Imperial times, see LSJ Suppl., s.v.) „(peasant / military) cloak (of coarse wool)" < Lat. *abolla* (Varro+; „origine inconnue" [Ernout – Meillet s.v., similarly Walde – Hofmann s.v.]).

3.4.3. *-ολλ- not < *-oli̯-.* In several other forms with surface -ολλ-, this sequence either does not derive from *-oli̯- or is not likely to have such a source, as in the following two categories:

(i) -ολλ- < *-oln-: despite the gemination (vs. expected lengthening, see e.g. Rix 1976: 67), a νυ-present seems assured for ὄλλυμι „destroy, kill" (cf. Harðarson 1993: 222f.)

(ii) -ολλ- from other sources:

(ii.1) Ἀπόλλων: the *e*-grade forms (WGk. Ἀπέλλων etc.) are older (see Meier-Brügger 1992: II.43, with references), thus Ἀπόλλων via vowel-harmony or otherwise secondary;

(ii.2) σκόλλυς (a type of haircut): hypocoristic gemination, cf. σκολ- in σκολύπτειν „pluck out" and similar forms, beside PN Σκόλλος.

3.4.4. *-ολλ- < *-oli̯- (?).* Most important for our purposes are forms with some chance of deriving from the sequence in question (and which otherwise should have ended up as -υλλ-); a source *-oli̯- is at least a possibility for the following three items with surface -ολλ-:

(i) κόλλα [Hdt.+] „glue" (cf. Hom. κολλήεις, κολλητός) : Slav. **klějĭ/*klĭjĭ* (CS *klěi* etc.) „id.", Du. *helen* (**haljan*) – Pr.-Gk. **koli̯ă* is traditionally interpreted in terms of a fem. devī-form **kol-ih$_2$* („dérivé f. en **yə$_2$*", Chantraine; similarly Frisk and others); but the form is perhaps rather an original collect./pl. **koli-h$_2$* („gluey stuff") to an *i*-stem **kóli-*.[63] In this case, the predicted phonologically regular *κύλλα (according to the above hypothesis) could have been restored after *i*-stem forms *κόλι(ς), *κόλιος etc., before these were ousted by the reinterpreted κόλλα.[64]

(ii) αἰόλλω [υ 27+] „move rapidly to and fro", αἰόλλομαι [Hes. *Sc.* 399+] „change color" – Even if formed early enough to have had the requisite sequence *-oli̯-, αἰόλλω is evidently based on the earlier-attested Hom. αἰόλος „fast-moving, shimmering",[65] so that a predicted *αἰύλλω would certainly have been restored to αἰόλλω (cf. ἄγγελος : ἀγγέλλω etc., Risch 1974: 286, Schwyzer 1939: 725).

(iii) πολλο- (: πολύς „much, many") – the source of the geminated stem-form πολλο- represents a complex and long-standing problem, for which a sequence *-oli̯- is only one of several possibilities; even if Szemerényi's widely-accepted account of this sort[66] turned out to be correct,[67] a *πυλλο- would have been easily restored to πολλο- after πολυ-.

There appear, then, to be no serious obstacles to the proposed rule *-oli̯- > *-ull-*, which can thus be used to explain the vocalism of at least φύλλον, μύλλω, σκύλλω, and κυλλός.[68] Nor should it be considered problematic that *-ori̯- shows no evidence of behaving in the same way, given probable phonetic differences between /r/ and /l/; it may be significant, in this connection, that epigraphic spellings with *i*-infection sometimes taken to be relics of the stage at which liquids became palatalized are quotable for /l/,[69] but not /r/, not to mention later dialectal palatalizations also quotable for /l/ (of the type Coan <ξυλεων> = /ksul'ōn/ etc., Buck 1955: § 71b).

4. **(-)TU̯OR-* (INCLUDING **SU̯OR-*)

4.1. We can now return to Schindler's *o*-grade interpretation of συρκ- (cf. 1.1.). If, as I have argued, a flanking labial and liquid (or nasal) are insufficient of themselves to trigger an **o* > υ process, two courses remain open: either we must abandon Schindler's interpretation and opt, with Lubotsky, for a zero grade (with everything implied thereby, namely „irregular *u*-colored zero grades" that surface under unclear circumstances); or we can explore other ways of justifying an *o*-grade. Given the manifest undesirability of having to resort to irregular *u*-coloration, we can reasonably pursue the latter course, first by considering, as with φύλλον, whether the broader

phonetic context might justify still another **o* > υ process. If we seem in danger of becoming unduly profligate with **o* > υ rules, it is sufficient to recall that Greek precisely has a problem with unexpected υ's; in principle, there is no difficulty with proposing that some Greek „irregular υ's" result from several distinct phonological sources. In the present case, I propose that **o* > υ when preceded by a *cluster* consisting of dental obstruent plus labial glide,[70] as in the following examples:

4.2. **(-)tu̯or-* > **(-)tu̯ur-*.

4.2.1. *συρκ-*. Cf. 1.1. above. Against Lubotsky's objections, one should note that the glossators' tradition labels this form as both Aeolic and Doric (*EM* 708.30ff. Gaisford), and not just Aeolic (more on this point below, 4.2.2.).

4.2.2. *Lesb. πέσσυρες* (cf. Hom. πίσυρες) „4". Lubotsky's account of Lesb. πέσσυρες (1.1.) is the traditional one; cf. the detailed study by García Ramón (1984), who establishes πέσσυρες (not πέσυρες) as the correct Lesbian form,[71] and who attempts to justify the alleged special Lesbian phonetic treatment of zero grade „**-tu̯^{o}r-*" (> **-tu̯ur-*) on the basis of the following two cases:

(i) συρκ- < **tu̯^{o}rḱ-* (cf. above);

(ii) σῦρω „draw, drag, sweep away" < **tu̯^{o}r-i̯e/o-*, beside the normal treatment of the zero grade in σαίρω „sweep" (cf. Chantraine s.v. 1 σαίρω: „Un seul fait évident, le rapport entre σαίρω et σῦρω".

It is immediately suspicious, however, that although this treatment is taken to be characteristic of Aeolic (thus also apparently Lubotsky), the two supporting examples, other than πέσσυρες itself, have a broader dialectal profile: cf. 4.2.1. for συρκ-; σῦρω is well-attested also in Attic and Ionic. If, however, **-tu̯or-* > **-tu̯ur-* regularly, Lesb. πέσσυρες can directly continue the old *o*-grade of **kʷetu̯óres* (Ved. *catvā́ras* etc.). As for the preservation of such a form in (at least part of) Aeolic (as opposed to the generalized zero grade of Attic-Ionic): whatever the explanation of the consonantism of WGk. τέτορες (on which see Lillo 1988, 1990: 14ff.), an old and still intact *o*-grade seems clearly to be needed for this form under any interpretation, and thus does not appear unthinkable also for Lesb. πέσσυρες.[72]

4.2.3. *σῦρω/σαίρω* (cf. 4.2.2.). According to the present hypothesis, σῦρω would continue an *o*-grade **tu̯or-i̯e/o-* (> **tu̯ur-i̯e/o-*), beside zero grade **tu̯r-i̯e/o-* in σαίρω. While the ultimate morphological interpretation is unfortunately hampered by the uncertain nature of the extra-Greek material,[73] a parallel **tu̯or-i̯e/o-/*tu̯r-i̯e/o-* might point to an original athematic *o/e*-verb of the *molō* type (cf. 3.3.1. above, and σκύλλω/σκάλλω, 3.3.2.).[74]

4.3. **dhu̯or-* > **dhu̯ur-*. For sequences with voiced aspirate, I briefly consider two inconclusive but suggestive cases.

4.3.1. *θύρσος „thyrsus"*. For the eastern cultural term θύρσος (i.e., the wand wreathed in ivy and vine-leaves topped with a pine cone, carried by worshippers of Dionysus), the traditional connection with HLuv. *tuwarsa-* „vine, vineyard" has seemed attractive.[75] If, however, the Greek form is simply borrowed (as seems to be widely assumed),[76] this would require, as seen already by Bossert, a hypothetical contracted variant HLuv. /tursa-/. While such a variant is unproblematic from the point of view of Luvian, it is disquieting, for the interpretation of Gk. θύρσος, that HLuv. /tursa/ is in fact never found, while HLuv. *tuwarsa-* itself is relatively well-attested (see Meriggi, loc. cit.). In theory, the **o* > υ process under discussion would allow for an inherited **dhu̯orso-* to result in both Gk. θύρσος (via Pr.-Gk. **tʰu̯urso*) and HLuv. *tuwarsa-* directly; but even if a plausible root etymology could be found,[77] such an approach faces other difficulties: the semantic and cultural profile of θύρσος surely favors borrowing as the more likely alternative, and inherited medial **-rs-* should not in any event yield pan-dialectal -ρσ- (see the extended discussion on this point by Forbes, loc. cit.). A more promising approach was outlined by Neumann, in his study of the signs for „wine" and „vine" in Aegean scripts (1977). Neumann identifies θύρσος and HLuv. *tuwarsa-* with the Linear A sign for „wine, vine" (1977: 128ff.), which he interprets phonetically as /tu/, indicating that the Minoan word it represents began with this sequence. Thus both θύρσος and *tuwarsa-* would be early independent borrowings from an eastern Mediterranean substrate language, processes in which Minoan may have played a role.[78] If this is the case, it seems possible that a source form (roughly) /tʰu̯orso-/ could have been borrowed early enough to result in both pre-Luvian /twarsa-/ or /dwarsa-/ and early Gk. /tʰu̯urso-/ (via the process in question), whence HLuv. *tuwarsa-* and Gk. [Eurip.+] θύρσος.

4.3.2. *θύραι „gate(s), door(s)"*. As is well known, the Greek *ā*-stem has replaced an original root noun with *o/ø* ablaut (not including several isolated athematic survivals, e.g. Hom. θύρᾰζε < **θύρᾰς-δε*; **θύρας* = Ved. acc. pl. *dúras*).[79] Although zero grade of the root is attested elsewhere (as prominently, for example, in Germanic and Baltic), the generalization of the zero grade in Greek is somewhat puzzling: thus Griepentrog (1995: 140) plausibly

explains the thematization of (Pr.-Gk.) **thur-* > **thur-ā* on the basis of its formal anomaly (there being no monosyllabic nouns in short vowel + /-r/), but does not account for the loss of the strong stem-forms **thu̯ōr-/*thu̯or-*, which is a necessary premise of his explanation. According to the above hypothesis (and with the additional assumption that /ō/ is treated like /o/), this would be easily explained by the inference that **thu̯ōr-* and **thu̯or-* would have fallen together phonologically with the weak stem **thur-*. It may also be suspicious that every one of the athematic relics has the shape θυρ-; while this is appropriate, as noted, for θύραζε, the θυρ- of other forms (i.e. Arc. θύρδα and θύσθεν[80]) could conceivably continue one or another of the strong stem-forms.[81] A potential problem with this sort of analysis concerns the fact that **-u̯-* after dental stops apparently remained intact until relatively late, in view of Homeric treatments of words with **(-)du̯-*, whereas Myc. *o-pi-tu-ra-jo* „door guard" (vel sim.), according to the hypothesized treatment, shows **-thu̯ur-* > **-thur-* already completed (as does the failure of θυρ- to make position in Homer). It does not seem difficult, however, to assume that **-u̯-* in this position was lost extremely early, which could be compared with other early instances of dissimilatory *u̯*-loss,[82] and more specifically with Perpillou's findings (1976: 49ff.) about the treatment of ϑ_F in epic.[83]

4.4. **su̯or-* > **su̯ur-*. Gk. ὕραξ „shrew, mouse" is evidently cognate with Lat. *sōrex* „id.": thus the latter would continue **su̯ōr-ak-*, while ὕραξ would seem to continue a zero-grade **sur-ak-*.[84] The parallelism in formation, however, is deceptive, and does not guarantee an inherited **su̯ōr-ak-/*sur-ak-*. It is doubtful that one could justify a class of inherited „*ak*-stems", whereas one certainly finds similar semantic profiles shown by semi-productive *ak*-formations in individual languages. Thus the Latin and Greek forms can easily be independent, as shown by similar material in both languages: compare, on the Latin side, other „Kleintiere" in *-ex*, some clearly secondary (*laurex* „young rabbit", *mūrex* „mollusk", *dentex* „sea bream", *cīmex* „bug", *culex* „gnat", *pūlex* „flea"; see Leumann 1977: 375, Ernout 1946: 142); and for Greek, note other animal terms in -ακ-, most of which are clearly secondary (Schwyzer 1939: 497, e.g. δέλφαξ „sow"), esp. „petits animaux" or „animaux considérés comme vils" (Chantraine 1933: 379: κόραξ „crow", πόρταξ „calf", σκύλαξ „puppy" etc.). Under these circumstances, there can be little certainty as to what type of inherited formation underlies the (probably or possibly einzelsprachlich) *ak*-forms (or at least *k*-forms) Lat. *sōrex* and Gk. ὕραξ. Nevertheless, the simplest assumption would be to begin with a root noun, with different apophonic

variants generalized in the two languages. In this event, an ablaut **su̯ōr-/*sur-* is distinctly unattractive. If, as is often suggested, the word belongs with the **su̯erH-* „surren etc." of Ved. *svárati* „makes a sound" etc. (thus e.g. IEW 1049), the root noun of such a set form should show zero grade throughout the paradigm, as Nussbaum has shown (1986: 66n53). As correctly observed by Schrijver, however (1991: 124), this connection is „no more than a possibility". Alternatively, if one could justify an anit version of the root (as in **su̯er-* „speak, answer", often assumed to be ultimately related to **su̯erH-* „surren etc."), this could reflect the agentive root noun type elucidated by Schindler (1972a,b; i.e. shrew = „squeaker, talker" vel sim.), which should show not **su̯ōr-/*sur-*, but rather **su̯ōr-/*su̯or-* in most of its paradigm. And for inherited root nouns referring to concrete objects and without clear verbal bases, zero grade is rare to the point of nonexistence. Thus, while there are no very strong positive indications, there is at least circumstantial evidence that Gk. ὕραξ could continue not a zero grade, but an *o*-grade **su̯or-ak-*.

4.5 *ὕλη „wood, woods"*. The appeal of the phonological proposal under discussion would be enhanced if it led to plausible etymologies for otherwise recalcitrant words; this may be the case with ὕλη [-ῡ-] „wood, woods" [Hom.+], an etymologically obscure (but evidently basic and old-looking) word.[85] The first step is to generalize **su̯or-* > **su̯ur-* (as may be seen in ὕραξ) so as to apply to **-ō-* and **-l-* (and not just **-ŏ-* and **-r-*; for /ō/, cf. on **thu̯ōr-*, 4.3.2.). On the semantic side, the present suggestion begins with Wackernagel's claim (1916: 185n1) that the original meaning of ὕλη is precisely „Brennholz", which he took to fit well with the fact that „für die ältere Zeit war das Wort bekanntlich ein Singulare tantum", and with his etymological connection with OIc. *usli* m. „glühende Asche", and ultimately Gk. εὕω „burn" etc. While this particular etymology (recently accepted by Leukart, 1994: 241n282) is far from compelling, for both phonological and morphological reasons,[86] Wackernagel's semantic interpretation is fairly attractive, certainly judging from the text of Homer, where ὕλη in the meaning „Brennholz" gives the impression of being old: it is not only well attested, but prominently found in passages referring to funeral preparations (including the funeral of Patroclus) and in contexts referring to religious offerings.[87] The same meaning is also implied by the only old compound of ὕλη, namely ὑλοτόμος.[88] These facts, together with the phonological treatment under consideration (in this case **su̯ōl-* > **su̯ūl-*), suggest a root identification with the **su̯el-* „burn, smolder" of OE *swelan* „kindle, inflame" (cf. Germ. *schwelen* „smolder, char", Li. *svilti* „smolder" etc.; IEW

2. *suel-*, 1045). As for semantics and morphology: ὕλη would reflect a **su̯ólā* meaning roughly „kindling" or „burning-material", either in prepared form („wood", e.g. as material for a funeral pyre), or in its natural state („woods"); formally, this would be a lengthened *o*-grade barytone feminine of the type [Hom.+] λώπη „mantle; covering, husk" < * „peeling(s)" (: λέπω „peel"), [Hes.+] ὥρα „care, concern, regard" (: ὁράω „see; take heed of"), κώρα· ὕβρις Hsch. (cf. κορέννυμι „be glutted"), and several similar forms (Schwyzer 1939: 459f.), a rare and clearly recessive type in Greek to which an isolated form like ὕλη could well belong.[89]

A minor problem concerns the shape of this root in Greek. The cognate usually cited is εἴλη or εἴλη „heat (of the sun)" [Aristoph. fr. 627+[90]], apparently with „prothetic" vowel (cf. Beekes' automatic laryngealistic reconstruction **h₁u̯el-*, 1969: 62). But Greek also preserves forms pointing to ϝελ-,[91] generally taken to show the expected „plain" **su̯el-*.[92] Both versions of the word are rare,[93] and it is perfectly possible that εἴλ-/εἴλ- reflects one or more secondary adjustments vis-à-vis ἔλ-/ϝελ-.[94]

4.6. *σορός*. The only counterexample to the process in question (for sequences with liquid after the *-o-*) appears to be σορός f. „funerary urn" (Hom., later „coffin"), at least according to the traditional interpretation in terms of a **tu̯or-ó-* „container", to be compared with Li. *tveriù / tvérti* „enclose; grasp, seize; form, fashion" (the latter meaning in Slav.: OCS *tvoriti* etc.), among other forms (IEW 2. *tu̯er-*, 1101), though none with certainty outside Balto-Slavic. Several considerations undermine the probative value of this form as a potential counterexample:

(i) σορός might only show that **Tu̯oR* > *TuuR* was restricted to (or at least favored in) accented syllables: the only exception would appear to be Lesb. πέσσυρες, which may be only apparent, since the recessive accent here is secondary.

(ii) σορός (according to the traditional etymology) belongs with a series of *e*-grade forms,[95] in which case the *o*-grade of σορός could have been maintained analogically, along the lines of 2.1. above.

(iii) The etymology (though recently cited with approval, for example, by Barton, 1993: 558) may itself be questioned, on several counts. If one must really begin with an Urbedeutung of the sort „umfassen" (rendered less than certain by the Slavic comparanda[96]), the specialized nature of the Greek term is at least suspicious, especially since Li. *tveriù / tvérti* means „enclose" mainly in the sense of „fence in" (a plot of land, etc.). Moreover, a Greek vessel term with feminine gender for an *o*-stem (and with a highly specialized

meaning) stands a certain chance of being borrowed (cf. ἀσάμινθος f. „tub" etc.), despite traditional attempts to ascribe the gender of σορός (and several other container-terms) to a simple sexual metaphor (e.g. Schwyzer - Debrunner 1950: 34).[97]

(iv) One could argue that the expected result of the process in question might actually appear in σύριχος / συρίσκος „basket", which, with its more general meaning, might be a better candidate than σορός for connection with Li. *tveriù* etc.; unfortunately, this word shows considerable formal variation that may again point to a non-native source, as in (iii),[98] or even some degree of contamination involving inherited and borrowed material, in which σορός may well have played some role. All in all, then, σορός cannot be considered a clear-cut counterexample to the process in question.

4.7. **Tu̯oN-*. Here I consider two possible cases of the same process in which the resonant after the *-o-* is a nasal.

4.7.1. *ὕμνος „hymn"*. The evidently old word ὕμνος [*Od.*+], generally glossed „hymn", is notoriously difficult etymologically. Frisk catalogues no fewer than six different approaches, to which a seventh can be added:

(i) cf. ὑμήν „membrane" („Band, Naht" > „Liedgefüge"), i.e. a thematized *men*-stem;

(ii) ~ (i), but based on the wedding-cry ὑμήν (possibly the same word as ὑμήν „membrane");

(iii) cf. ὑφή „weaving, web", ὑφαίνω „weave" (and usages like ὕμνον ὑφαίνειν);

(iv) cf. ὑδέω „besingen" (*ὕδ-μος, whence *ὕν-μος > ὕμνος regularly);

(v) cf. Ved. *sumná-* n. „benevolence, favor", also „prayer (?)", thematized from an agent noun **h₁su-mḗn* „one who thinks favorably" (see Peters 1980: 207, with references);

(vi) Mediterranean loan, cf. similar terms for sung compositions like διθύραμβος „dithyramb", ἔλεγος „lament", λίνος „song" (also personified), etc. (with ending influenced by λίνος and/or θρῆνος „dirge"?);

(vii) cf. Hitt. *išhamai-* „song", i.e. ὕμνος < **sh₂óm-no-* (Eichner apud Peters 1980: 210).

Despite occasional claims to the contrary,[99] none of these is compelling; their various difficulties need not be signaled in exhaustive detail.[100] In one respect, however, several of these approaches – i.e. those which, like (iv) and (vii), begin from a basic meaning „song" or „singing" – are semantically attractive, for an important reason that has gone generally unacknowledged. Although the lexica stress the meaning „hymn" of

ὕμνος,[101] the earliest attestation of the word shows no such sense: ὕμνος is attested only once in the *Odyssey*, with reference to the song sung by Demodocus concerning the Trojan horse, in a phrase ἀοιδῆς ὕμνον ἀκούων# (Θ 429).[102] The early epic usage simply shows that the word began with a general meaning like „singing", as is also clear from Hesiod, *Op.* 657 (ὕμνῳ νικήσαντα[103]) and 662 (Μοῦσαι γάρ μ' ἐδίδαξαν ἀθέσφατον ὕμνον ἀείδειν; cf. West 1978: 321 ad 657: „the word is not yet specialized in the sense 'hymn'", citing also h.Ap. 161 ὕμνον ἀείδουσιν, θέλγουσι δὲ φῦλ' ἀνθρώπων). The same feature is attested for early usages of the denominative ὑμνέω (e.g. Hes. *Th.* 70 ὑμνεύσαις, referring to the singing of the Graces and Desire).[104] Given, then, the early general sense „singing, song", I suggest that ὕμνος could continue an *o*-grade action-noun **su̯on[H]-mo-* *„sounding, intoning" > „singing, song" (to the **su̯enH-* „sound, intone" of Lat. *sonāre*, Ved. *ásvanīt* etc.), which, with laryngeal loss and **-nm-* > **-mn-* metathesis (both phonologically regular) and with the **Tu̯oN-* > **Tu̯uN-* treatment under discussion, would give ὕμνος directly.[105]

4.7.2. *Hom.* + *-οσύνη*. The Greek abstract suffix generally cited in the form -σύνη[106] has been considered to be related to the similar suffix Ved. *-tvaná-*[107] at least since Aufrecht 1852, and the comparison continues to be widely accepted.[108] The terms of the comparison (where these are acknowledged) generally follow Benveniste's morphological conception[109] via parallel thematizations of full-grade and zero-grade allomorphs (**-u̯en-o-/*-un-eh₂*) of primary *-u̯er/n-*stems, with the addition of the *-t-* proper to forms like Ved. *sú-t-van-* „pressing". The alternative inner-Greek explanation championed by Wyss (1954) continues to attract some partisans (recently e.g. Powell 1988: 105f., Ruijgh 1992: 94n60), but must be considered highly unlikely.[110]

The superficial attractiveness of Benveniste's model masks numerous problems and indeterminacies, the most serious of which is the fact that both -σύνη and *-tvaná-* behave as secondary suffixes, with -σύνη in fact further generalized in its form after *o*-stems (thus for Greek, the suffix is actually „-οσύνη"), while Ved. *-tvaná-* appears indiscriminately with *i*-stems (e.g. *kavitvaná-* „wisdom, visionary power"), *u*-stems (*vasutvaná-*), *n*-stems (*vr̥satvaná-* „manly power") and *o*-stems (*martyatvaná-* „mortality"), with the best-attested form (*mahitvaná-*, RV 14x) descriptively based on the adj. (neut.) *máhi*. Even though *-u̯er/n-*stems are known to behave in part as secondary formations, it is exceedingly difficult to construct a credible scenario whereby the primary **-t-* of forms like *sú-t-van-* can have become attached to what has ended up as (two thematized versions of) a secondary

suffix (the thematizations themselves being, to all appearances, semantically vacuous, although the function of secondary *-u̯er/n-* remains somewhat difficult to specify). Indeed, the complete lack of word-equations, as well as the formal features of the suffixes in each language, imply a long period of development (according to the theory that they are related) that is simply unrecoverable. Among other problems might also be noted the fact that -σύνη, under this analysis, presupposes a pandialectal assibilation of **-tu-* in Greek, which is difficult to accept as such.[111] In a recent discussion of Greek palatalization processes, Viredaz speculates (1993: 333 n. 8) that „[l']assibilation apparemment panhéllénique des suffixes -σις et -συν-o/ā- est peut-être un fait d'emprunt interdialectal d'époque mycénienne". It may indeed be attractive to assume that -σύνη, at least, is essentially an epicism.

A more serious problem arises from G. Banti's demonstration (1984: 94ff.) that Ved. *-tvaná-* may well be a secondary agglutination based on Ved. *-tvá-*,[112] specifically involving „distracted" instr. sg. forms like *mahitvā́* (32x, beside 19x acc.) extended to *mahitvanā́* (9x, beside 5x acc.), partly on the basis of pronominal forms like instr. sg. *enā́, amúnā*.[113] Although Banti makes no mention of -σύνη, his account, if correct, utterly explodes any direct connection between it and Ved. *-tvaná-;* and in that event we are free to explore alternative methods of accounting for Gk. -σύνη. To begin with, a theory involving an agglutinated IE **-tu̯o-* (whatever the ultimate analysis of this formant) would be a highly favored one, in view of similar patterns elsewhere (cf. Slav. *-ĭstvo, -ĭstvĭje*, Go. *waurstw* „work" < **urǵ-s-tu̯o-*). If, then, we begin with a sequence **-tu̯onā* for Gk. -σύνη,[114] we arrive precisely at the o > υ development under consideration.[115] To consider the generalized (post-*o*-stem) version **-otu̯onā* (cf. Hom. -οσύνη), the process in question will regularly yield (South Gk.) **-ossunā;* all that is needed is a method of accounting for the apparent degemination resulting in -οσύνη. If it is assumed that these abstracts in **-ossunā* were highly favored in epic language (as a result of nuances not readily recoverable to us), the Kunstsprache itself would have seen to the elimination of the unusable cretic pattern in a virtually automatic way, with recourse to the degemination (essentially a metrical shortening strategy) well known from patterns of the sort ἔπεσι / ἔπεσσι, Ὀδυσεύς / Ὀδυσσεύς, μαχέσασθαι / μαχέσσασθαι, etc. (Chantraine 1958: 179f.), with the difference that, unlike ἔπεσσι etc., geminated forms like †ἱπποσσύνη (-ῃ, -άων etc.) with internal cretic could under no circumstances occur (and in fact do not). For a partial parallel to this process, compare what appears to be a metrical shortening in φοινῐκόεσσαν# K 133+ (vs. etymological φοινῖκ-), evidently a relatively late replacement of

a *φοινῑκ-ϝεντ-; although such a formation would have been well-formed in Mycenaean times (cf. Myc. /ped-went-/ etc.), the generalization of „-οϝεντ-" (cf. above on -οσύνη) would have required a †φοινῑκ-οϝεντ-, which, in view of its unusable internal cretic, had to be adjusted (in this case by a secondary metrical shortening involving vowel quantity).[116] Certainly there are problems with this account, which unfortunately cannot be pursued in more detail here.[117] Nevertheless, the application of *-*tu̯on-* > *-*tu̯un-* to -σύνη allows for a potentially attractive interpretation that aligns -σύνη and Ved. *-tvaná-* as independent (but formally parallel) creations based on *-*tu̯o-*, which may after all be required by the Vedic facts, as noted above.

4.8 *σομφός*. The sole counterexample to the suggested development *$T\underset{\frown}{u}oN$- > *$T\underset{\frown}{u}uuN$- appears to be σομφός [Hippoc.+] „spongy, porous" (cf. 2.7.), under the traditional assumption that this reflects a *$s\underset{\frown}{u}ombho$- cognate with Gmc. forms like Eng. *swamp*. The initial σ-, however, would be of the same irregular type as in Dunkel's account of σύν (2.7.), and moreover the Greek Auslaut does not match that of Germanic; nor can σομφός be easily separated from σπόγγος / σφόγγος „sponge". In sum, we find variation and irregularity characteristic of a borrowing or (cf. Klimov's comparison of Kartvelian material [1991, 1993]) a Wanderwort; as recently emphasized, for these reasons, by Ringe (1995: 61n8), „this etymon *certainly should not* be reconstructed for PIE". Under these circumstances, σομφός has only minimal (and perhaps negligible) force as a counterexample to *$T\underset{\frown}{u}oN$- > *$T\underset{\frown}{u}uuN$-.

5. *$T\underset{\frown}{U}OR$- (> $T\underset{\frown}{U}UR$-) VS. *$C\underset{\frown}{U}O$- (> $C\underset{\frown}{U}U$) AND ὕπνος/ὕπαρ

5.1. Section 4. has sought to vindicate Schindler's account of Gk. συρκ- < *o*-grade *$t\underset{\frown}{u}ork$- (1.1.), by suggesting that *$T\underset{\frown}{u}oR$- > $T\underset{\frown}{u}uuR$- is a regular phonological process of Greek, visible in at least several comparable forms. There is, however, no phonetic basis for a restriction whereby the cluster preceding the *o* must begin with a dental obstruent (as in the notation *$T\underset{\frown}{u}oR$- used informally above): in principle, the same *o > u effect (perhaps phonetically a pursing effect similar to that in *$nok^{\underset{\frown}{u}}$ > /$nuk^{\underset{\frown}{u}}$/) might be expected after velar (*[-]$k\underset{\frown}{u}$-, *[-]$g\underset{\frown}{u}$- etc.) or labial (*[-]$p\underset{\frown}{u}$-, *[-]$b\underset{\frown}{u}$- etc.). Although sequences of the latter sort do not appear to occur, a rare form with velar of the type *$\acute{k}\underset{\frown}{u}\acute{o}n$-$\underset{\circ}{n}$ „dog" (acc. sg.) might thus have resulted directly (or almost directly, cf. 2.5.) in κύνα, according to this more relaxed formulation of the rule. In this event, the frequency of forms with dental obstruent preceding the glide would be the result of chance, which would not in itself be surprising, given the phonotactic properties of Greek. In the same way, one could question the apparent restriction whereby the segment

following the *o must be a resonant: in the absence of any clear phonetic basis for such a restriction, one should thus consider the maximally general formulation suggested in the heading of this section, according to which *o becomes υ following any cluster consisting of obstruent plus labial glide (regardless of the nature of the following consonant).[118] This conception has potentially interesting consequences for Gk. ὕπνος and ὕπαρ, with which I conclude.

5.2. *ὕπνος „sleep"*. This form has inevitably been set up as **sup-no-*, and compared with zero-grade forms in Slavic (OCS *sŭnŭ*) and Albanian (*gjumë*). In traditional accounts, forms pointing to a **sop-no-* in Celtic and Baltic have also been taken to provide indirect evidence for an old zero-grade **sup-no-*. In his study of words for „sleep" in Indo-European (1966), Schindler sought to revise some of the details of the analysis of the Celtic and Baltic material, but he did not dispute the existence of an old **sup-no-*, even attempting to add Pāli *supina-* as indirect testimony to the antiquity of such a form in Indic (although this example turns out to be specious[119]). Nevertheless, zero-grade **sup-no-* occurs beside *e*-grade *$s\underset{\frown}{u}\acute{e}p$-*no-* (Lat. *somnus*) and *o*-grade *$s\underset{\frown}{u}\acute{o}p$-*no-* (Arm. *k'own*), the latter two reflecting a type of deradical *-*no-* formation attested in such forms as *$\underset{\frown}{u}\acute{e}gh$-*no-*/*$\underset{\frown}{u}\acute{o}gh$-*no-* „wagon" (: *$\underset{\frown}{u}eǵh$- „convey"), or *$\underset{\frown}{u}\acute{e}s$-*no-*/*$\underset{\frown}{u}\acute{o}s$-*no-* „price" (: *$\underset{\frown}{u}es$- „buy").[120] Schindler (1966: 75n28) explicitly rejected various attempts to explain the zero-grade vocalism **sup-no-* secondarily,[121] suggesting, first, that all three ablaut grades could be explained on the basis of a primordial *r/n*-stem *$s\underset{\frown}{u}\acute{e}p\bar{o}r$ (gen.**supnés*, loc. $s\underset{\frown}{u}op$-*én*), and second, that *$\underset{\frown}{u}\acute{e}s$-*no-*/*$\underset{\frown}{u}\acute{o}s$-*no-* provided a good parallel to **sup-no-* (beside *$s\underset{\frown}{u}\acute{e}p$-*no-*/*$s\underset{\frown}{u}\acute{o}p$-*no-*), insofar as a zero-grade **us-no-* was implied by Hitt. *uš(ša)ni$\underset{\frown}{i}a$-* „sell". Watkins (1972: 558), arguing against Schindler's conception of an *r/n*-stem, suggested that such forms „probably represent extensions in *-*no-* to old apophonic root nouns". However the ablaut variation is to be explained, a certain importance clearly attaches to the status of the zero-grade **us-no-*, allegedly parallel to **sup-no-*.

In fact, the existence of a Hitt. *uš(ša)na- underlying *uš(ša)ni$\underset{\frown}{i}a$-* has been questioned by Neu (1980: 87f.), who notes that a Hittite denominative like *uš(ša)ni$\underset{\frown}{i}a$-* could just as well have been based on a full-grade *$\underset{\frown}{u}\acute{e}s$-*no-* or *$\underset{\frown}{u}\acute{o}s$-*no-* (> Hitt. **wašna-*), which could have been regularly reduced in an originally end-stressed formation **us-no-$i\acute{e}/\acute{o}$-*. Barton (loc. cit.) lends confirmation to Neu's observation by providing a more explicit account of this formation that dispenses entirely with zero grade.[122] If, then, zero grade is expendable for this type, and also in view of the *o*-grade of Arm.

k'own, Gk. ὕπνος could in principle continue **su̯óp-no-* directly, under the assumption of the more general *Cu̯o* > *Cuu* rule under consideration. In the end, however, the case is indeterminate, given possible external evidence (as noted) for a zero-grade **sup-no-*.

5.3. *ὕπαρ „(true) dream"*. Unlike ὕπνος, Gk. ὕπαρ [*Od.*+; no other form in use, and ὕπαρ partly adverbial] is isolated as a free-standing nominal form: the cognate *r*-stem forms Hitt. *šuppariia-* „sleep" and OIr. *tindabrath* „falling asleep" (Watkins 1976) are essentially denominative verbs (or more precisely, in the case of *tindabrath*, a verbal noun based on an unattested denominative verb). Thus, although Hitt. *šuppariia-* is routinely thought to be based on a noun **šuppar* assumed to be identical to Gk. ὕπαρ, both reflecting a zero-grade neuter *r*-stem **súp-r̥* (see Watkins, 1972: 556, 558), this need not be the case. Just as the denominative Hitt. *uš(ša)niia-* does not exclude the existence of a full-grade **wašna-*, the denominative *šuppariia-* can also in theory be based on a full-grade **šuu̯appar* (reduced to **šuppar-* in an end-stressed denominative in **-i̯é/ó-*).[123] Similarly, what Watkins describes as the „morpheme-isolate" *-abr-* „sleep" (NB: not „dream") in OIr. *tindabrath* is attested only in composition, and can continue a derivative **-sup-rV-*, in principle a thematic **sup-ro-* „sleep" based on (but not necessarily identical to) the athematic *r*-stem basis.

We have already noted (5.2.) Watkins's claim that the construct of a **su̯épōr/*supnés*, for which Schindler argued in his 1966 paper, has no real basis. Indeed, Schindler's later (and by now almost canonic) conception of the ablaut of *r/n*-stems (1975) would have viewed such a form quite differently, although his paper did not take up this word. According to this later framework, zero grade of the root is appropriate only to the following types of forms:

(i) weak cases of what Schindler then called the „holokinetic" collective, of the type **u̯éd-ōr/*ud-n-* „water" (whence the generalized zero grade of forms like ὕδωρ, σκῶρ „excrement");

(ii) renewed zero grades in the weak cases of Schindler's acrostatic „type I" (**u̯ód-r̥*; gen. ***u̯éd-n̥-s* > **ud-én-s/*ud-n-és*), cf. his discussion (1975: 7f.) of the zero grade stem seen in Ved. *ū́dhar* „udder";

(iii) weak cases of proterokinetic „complex" *r/n*-stems (esp. *u̯er/n*-stems), cf. the generalized zero grade in Gk. πῖαρ „fat".

An archaic noun **súp-r̥* seems very difficult to explain within this framework, particularly in terms of (i) and (iii): there is no evidence that the form is a collective (Gk. ὕπαρ, not †ὕπωρ), and type (iii) („complex" *r/n*-stem)

is simply inapplicable. There is little justification for setting up a basic type of zero-grade *r/n*-stem,[124] and even if the forms pointing to a Slav. **sŭpar-/*sŭpan-* „sleep" reflect an archaic pattern in Slavic (Ivanov 1991: 27ff.: cf. Slav. *dar-/dan-* „gift"), the suffix formation (as Ivanov points out) is phonologically ambiguous, and in any case need not point to an *inherited* heteroclitic. There remain, then, two possibilities for Gk. ὕπαρ[125]:

(a) ὕπαρ = generalized zero grade of an acrostatic „type I" noun, i.e. **su̯óp-r̥/**su̯ép-n̥-s* > **sup-én-s/*sup-n-és* (pattern (ii) above), whence generalized zero grade in ὕπαρ (cf. Ved. *ū́dhar*);

(b) ὕπαρ = the acrostatic „type I" **su̯óp-r̥* itself, via the *Cu̯o* > *Cuu* process under discussion, with ὕπαρ thus directly comparable to the Hitt. **šuu̯appar* (cf. *u̯atar, šakkar, lammar, pattar* etc.) postulated above.[126] In view of the *o*-grade of Gk. οὖθαρ (vs. Ved. *ū́dhar*) and the archaic appearance of the frozen form ὕπαρ, the second of these is the simpler alternative, and (barring an isolated zero-grade *r*-stem of the type κύαρ, which cannot perhaps be absolutely excluded) seems preferable.

5.4. *ὥς „like, as"; σῶος „safe and sound"*. Finally, I note two potential counterexamples to the more general formulation of this process (cf. 5.1.): these forms, in their traditional interpretations, would call into question the regular development of **su̯ō-/*tu̯ō-* to **su̯ū-/*tu̯ū-* (cf. **su̯ōl-* > **su̯ūl-* in ὕλη, 4.5.), but neither seems to me probative:

5.4.1. The comparative particle Hom. ὥς „like, as" is traditionally said to be cognate with Go. *swa* „thus, (so) that" and *swe* „when, as" (Frisk s.v. 3. ὥς , Lehmann - Feist 331f. s.v. *swa*), thus reflecting a **su̯ō+s;* but it is equally possible (and perhaps preferable) to interpret the Greek form as continuing plain **u̯ō+s*, as argued by Dunkel (1982/1983: 193; 1988: 119, following Wackernagel) on comparative grounds (cf. Skt. *iva* etc.). An additional argument in favor of this account may come from the particle's Homeric usage, which is not merely postpositive but also typically line-final (i.e. of the type Γ 2 ὄρνιθες ὥς#, Z 443 κακὸς ὥς# etc., cf. Chantraine 1958: 126, Hoenigswald 1991: 8): ὥς in this usage behaves identically, as Hoenigswald shows, to other line-final monosyllables beginning with a single consonant (νύξ#, πῦρ# etc., virtually always preceded by diphthong, or by short vowel plus single consonant), but fails to match the behavior of the reflexive pronoun (cf. the type θυγατέρα ἥν# E 371+ [Hoenigswald 1991:4], i.e. **ϝϝήν < **su̯-*). Thus the failure of ὥς to make position (identical in this respect to (ϝ)ἵς#) would be most easily explained by **u̯ō+s*.

5.4.2. σῶος „safe and sound" (cf. σάος „id.") has been claimed to show an old full-grade *$t\underset{\frown}{u}oh_2$-$\underset{\frown}{u}o$-, beside the irregularly syllabified *$t\underset{\frown}{u}ə_2$-$\underset{\frown}{u}o$- of σάος (thus e.g. Lejeune 1972: § 100); in this case, *$t\underset{\frown}{u}oh_2$-$\underset{\frown}{u}o$- would produce a Pr.-Gk. *$t\underset{\frown}{u}\bar{o}\underset{\frown}{u}o$- that does not end up as *$t\underset{\frown}{u}\bar{u}\underset{\frown}{u}o$-. This is, however, a very problematic analysis, not least of all because of the difficult Schwebeablaut it would require if one keeps the otherwise attractive connection with the *$teuH$-/*tuH- „swell, be strong" of Ved. *távīti* „is strong" etc., as is generally done even under this analysis.[127] σῶος is better taken as a late thematization of σῶς, itself contracted from σάος (thus e.g. Beekes 1969: 249, after Leumann).[128]

6. CONCLUSIONS

The results of the foregoing discussion may be summarized as follows:

(i) A Cowgill's Law „proper" can be said to account for two types of forms:

(a) *o > υ in /noK$^{\underset{\frown}{u}}$, K$^{\underset{\frown}{u}}$on/ (νύξ, γυμνός; ὄνυξ if with labiovelar, γυνή if < o-grade); similarly /g$^{\underset{\frown}{uo}}$n̥(n)-/ > /gun-/ (γυνή if < zero grade),[129] /noKh$\underset{\frown}{u}$/ > /nuKh($\underset{\frown}{u}$)/ (ὄνυξ if without labiovelar);

(b) *o > υ in /nom/ (ὄνυμα, -νῦμ- in nasal presents < *-n-h_3+m-), but not in /mon/ (cf. μόνος).

(ii) There is little or no evidence for *o > υ in the following types of cases:

(a) the (Cowgill / Sihler et al.) „generalized" version of Cowgill's Law (*o > υ between labial and resonant);[130]

(b) forms with intervening liquid (e.g. πρυμνός, μύρμηξ);

(c) /soN/ (σύν).

(iii) Two additional *o > υ processes (not necessarily related to [i]) appear to be needed for Greek:

(a) *-oli- > -υλλ- (φύλλον, μύλλω, σκύλλω, κυλλός);

(b) *(-)$T\underset{\frown}{u}oR$- > *(-)$T\underset{\frown}{u}uR$- (συρκ-, πέσσυρες / πίσυρες, σῦρω; perhaps κύνα, θύρσος, θύραι, ὕραξ, -οσύνη and [via new etymologies] ὕλη, ὕμνος).[131]

(iv) If (iii.b) is to be stated in more general form (*$C\underset{\frown}{u}o$- > *$C\underset{\frown}{u}u$-), ὕπνος and/or ὕπαρ could continue o-grades.

(v) Most generally, the above analyses, if broadly accurate, have the highly desirable morphological consequence of eliminating altogether the problematic notion of an unconstrained „alternate Schwundstufe", or irregular u-colored prop vowel treatments (as in traditional Greek „u-grades" of the sort *$bh^{\underset{\frown}{u}}\underset{\frown}{l}io$-, *$t\underset{\frown}{u}^{\underset{\frown}{u}}\underset{\frown}{r}ie/o$- etc.).[132]

NOTES

1. Hamp follows Forssman's reconstruction (1965).

2. Forssman, in his discussion of the „fakultativer Wandel o > u in Umgebung von Labialen, besonders im Lesbischen" (1965: 14f.), considers on an equal footing the late dialectally-restricted phenomena reflected by forms like Lesb. ὕμως (Att. ὅμως), ὑπίσσω (Att. ὀπίσσω) etc., as well as the early pan-dialectal type represented by νύξ etc.; certainly on the grounds of chronology, however, these two must be carefully distinguished, and it is the latter that is the primary focus of this investigation.

3. Cf. Lejeune (1972: § 192 n. 3): „De manière générale, là où il ne répond pas à un *$\breve{u}$ indo-européen, le u du grec est souvent d'interprétation difficile". On possible Medieval and Modern Greek analogues to some of the Ancient Greek „irregular υ" data, see Joseph 1979.

4. E.g. Schwyzer (1939: 351): „ρυ (λυ) bzw. υρ υλ stehen in einer Anzahl von Fällen für idg. $\underset{\frown}{r}$, $\underset{\frown}{l}$" (citing φύλλον and μύλη, among other forms; earlier references on p. 352 – note esp. Güntert 1916: 32ff., § III. [„Zur u-Färbung reduzierter Vokale im Griechischen"]).

5. In these terms e.g. Lejeune (1972: §211 nI citing φύλλον and μύλη).

6. I leave aside several more radical attempts to explain Greek „irregular υ", such as Sapir's early laryngealistic approach (1938: 274), no longer tenable, or Meillet's suggestion (easily falsifiable) that *e and *o before syllable-final labiovelar regularly yield Gk. υ (1896: 136).

7. For γυνή via *$g^{\underset{\frown}{u}}en$- > *$g^{\underset{\frown}{u}}on$-: Cowgill 1965: 157n33 (following Szemerényi).

8. See e.g. Mayrhofer 1986: 136 and EWAia II, 82 (with further references).

9. Although „Frauensprache" may well be involved, I cannot accept the details of Knobloch's speculative account (1993); for recent discussion of the formal particulars, see also Beekes 1994: 91ff.

10. Cf. Frisk (s.v. νέμω): „Die weitverzweigten Bedeutungen von νέμω nebst Ableitungen bieten ein Problem, das kaum endgültig gelöst ist".

11. On the need for this method of accounting for Gk. νυ-presents see also Sihler 1988: 555, and more recently Harðarson 1993: 222f.

12. Despite e.g. Pokorny (IEW 749), it is far from certain that Μυρμιδόνες belongs here as „das Volk der Schreckgespenster".

13. On this pattern see now also Klein (1996: 38), with comments on *to- (particle of „du deixis") beside *$tu(H)$ (2sg. pron.).

14. See recently on this point EWAia II, 507 (s.v. *vamrá*-), and earlier e.g. Pisani 1966: 235 („Tabuisierung").

15. For possible cases of *u*-vocalism outside Greek, note Ernout – Meillet (s.v. *formīca*) on the Lat. variant *furmīca* („Le *-ur-* de *furmīca* peut donc etre ancien"), as well as Meillet 1935 on Arm. *mrǰiwn,* which, despite Pisani (1966: 227), could well reflect earlier **murǰiwn,* whatever the best interpretation of such a form might be.

16. The effect of these sequences is relatively easy to motivate phonetically, given not only the increased rounding naturally associated with the pursing effect of the labiovelar (and perhaps similarly with a tense rounded /m/), but also the convergence in formant structure between nasals and high vowels.

17. This section has benefited from discussion with Leroy Brockman.

18. Beekes adduces a second factor to account for the *u*-vocalism, namely a dissimilation o…o > o…υ, comparing ω…υ > ω…υ and compounds *-ῶνυξ/-ώνυχος (cf. 2.2. on -ωνυμ(ν)ος, probably already Mycenaean (*re-u-ko-nu-ka,* Meier-Brügger 1992: II, 38).

19. The general problem was evident to Szemerényi (1964: 240n4), although he stopped short of setting up **h₃nog^{u}h-* for ὄνυξ; cf. more recently Ringe 1988: 431n44 („Positing a labiovelar … makes it easier to explain the *u* of Gk. *ónuks*"). A labiovelar interpretation might also be preferable for Hitt. *šankuuai-,* thus pl. *šankuuai* = collect. **[s+]h₃n̥g^{u}h-ōi* with generalized zero grade (type Gk. ὕδωρ „water", σκῶρ „excrement"; for unstressed *n̥C* in Hitt. see Melchert 1994: 125f.), vs. **[s+]h₃enghuōi* (collect. type **uédōr,* but requiring [*i*-extended] *u*-stem and Schwebeablaut), on which see Eichner 1985: 165 and Harðarson 1987: 80f., 88.

20. An account of this sort seems assumed by Ringe (1996: 99).

21. For discussion and references see Melchert 1994: 49ff. Jochem Schindler's catalytic role in promoting interest in this phenomenon – evidently on both sides of the Atlantic (cf. Eichner 1988: 140n58) – was not the least of his achievements.

22. For κύκλος (and Myc. *ku-ke-re-u*) in these terms with old *e*-grade: recently Plath 1994: 240n33. Rounding of */e/* by flanking labiovelars is of course typologically trivial; within IE, and ultimately involving the same word as κύκλος, cf. Toch. B *kokale* „chariot" and similar forms (recent discussion of this rounding process in Katz 1994: 160f., somewhat differently Ringe 1996: 144).

23. Eichner 1985: 134ff. (esp. 139n31) and apud Mayrhofer 1986: 176n334, cf. Oettinger 1993: 212, with further references.

24. For (i) and (ii) see Frisk s.vv.

25. Similarly Hamp (1970a: 486), who finds the intermediate stage „**g^{u}onā*" to be „not well-motivated".

26. On the original paradigm of „woman", with somewhat more complex assumptions about the IE preform, see Jasanoff 1989; additional recent bibliography in de Lamberterie 1991: 153n19.

27. Somewhat differently de Lamberterie (1991: 154f.), assuming zero grade generalized throughout („nom. sg. γυνή (béot. βανά) < i.e. **g^{wo}nā*"), although he provides other evidence outside this paradigm for the survival of the *e*-grade in Greek.

28. The problem posed by this discrepancy has been addressed in explicit fashion by Carruba (1991: 166, 179n28), who nevertheless adopts the extreme (and clearly unacceptable) solution of interpreting the υ of γυνή as deriving from inherited **u*.

29. These forms, strictly speaking, are ambiguous between *o*-grade (the simplest „face-value" interpretation) and zero grade (for discussion, see Melchert 1994: 264, 332).

30. The Cypr. βονᾶ „wife" sometimes seen in the *(-)po-na* of the obscure text DGE 685.2 = ICS § 298 is far too uncertain to be of value here (see ICS ad loc.).

31. See recently Knobloch 1993: 303 (with reference to his earlier treatment); **um(e)no-* „ein Leitsuffix für mediterrane Wörter".

32. Lesb. ὐσδ- is in any case quite possibly specious (cf. Blümel 1982: 45n21, following Nickau 1974).

33. Perhaps ultimately with Ved. *sám,* but see the reference to Tichy at EWAia II, 704.

34. See the references (other than Dunkel) at Meier-Brügger 1992: I,156, as well as Hamp 1975: 470.

35. E.g. λοϝ- „wash", not †λυϝ- (but Myc. /lewo-/); ἑφολκός „attractive", not †ἑφυλκ- (but /ep(i)-holk-/ : ἕλκω „drag"); the type μολεῖν „come, go", not †μυλ- (but cf. βλώσκω „id."); various *e/o*-ablaut types, e.g. ῥόπαλον „club", not †ῥυπ- (: ῥέπω „sink, fall"), μολπή „song", not †μυλπ- (: μέλπω „sing"), etc.

36. E.g. μόλυβδος / μόλιβος / βόλιμος „lead" (but very early, cf. Myc. *mo-ri-wo-do*), μολόχη/μαλάχη „mallow", ῥομφαία „(Thracian) broad sword", etc.

37. E.g. λοφνίς „torch (of bark)" (< *λοπ-σν-ο-), not †λυφ- (but belongs ultimately with λέπω „peel"); κόλπος „lap, fold etc." (< **kᵘolp-* with dissimilatory *u*-loss), not †κ(ϝ)υλπ- (adjust relative chronology, i.e. order *u*-loss before **o* > υ); πόλις „city", not †πυλ- (appeal to variant πτολ-, and/or the possibility of a zero-grade source **pl̥h₁*); πολιός „grey" (Myc. *po-ri-wa*) (cf. related *e*-grade πελλός „dark"); etc.

38. With regular laryngeal loss by „Saussure's Law".

39. Even if forms like Myc. *o-ro-me-no* and Hom. ὄρονται suggest that a root **ser-* lies behind some (or even most) of this material (see e.g. Leukart 1994: 93f. and Hackstein 1995: 262ff. for recent discussion), **u̯er-* is assured at least for impf. ἐώρων (cf. also Myc. *Da-i-wo-wo /Da^{h}i-worwos/* 'der im Kampf wachsam ist', *E-u-ru-wo-ta /Euru-wortās/* 'der weit blickt, späht', Leukart 1994: 52n13, 93).

40. I.e., under the likely assumption that synchronic association with related (but not derivationally related) *e*-grades (e.g. εἰλέω [Soph.+] „wind, turn", cf. ἀνεείλε [h. Merc. 239, ex coni.], etc.) would not have been sufficient to prevent application of the alleged **o* > υ process.

41. Probably also Hom. μορόεις, see the discussion by Janko (1992: 177f.).

42. Even if μέλας „black" ultimately belongs here etymologically, this should not have prevented **o* > *u*.

43. The variant ῥυφ- of ῥυφέω / ῥυφάνω is Ionic only (cf. my comments on Ion. ὑπεατ- beside Att. ὄπεαρ / ὀπηρ-, 1994: 32). It is thus unnecessary to claim (as e.g. Schwyzer 1939: 351) that ῥυφ- continues the zero grade *$\overset{\circ}{s}rbh$- of Lat. *sorbeō*.

44. His arguments are based in part on the suggestive fact that as a technical term, *$melH$- „grind" has been largely replaced by *h_2elh_1- in both Greek (ἀλέω) and Armenian (*alam*). (This particular Greek/Armenian lexical parallel is not significantly weakened by Clackson's strictures [1994: 90ff.] against certain morphological parallels often said to hold between Greek and Armenian material based on *h_2elh_1-).

45. See Hamp 1991 on σπυρίς „basket, creel" and ἄγυρις „gathering, crowd", in addition to μύλη. (Other well-known candidates with υ include πύργος „tower" and τύμβος „burial mound, sepulchre"; Cowgill 1986: 60, Clackson 1994: 121). I concur with Hamp's judgment (1991) that Rasmussen's attempt to explain μύλη and similar forms via deviant or otherwise unattested IE root structures of the sort *$mwelH$- (1987) is „ingenious but scarcely persuasive" (cf. next footnote).

46. Cf. Leumann 1977: 44 („lat. *o* ererbt" in *folium*). Again, I cannot accept Rasmussen's *$bhwel$- (1987: 40f.).

47. Note that, according to this formulation, the presence or absence of a labial (before the sequence *-*oli̯*-) is taken to be irrelevant.

48. It is apparent that the following discussion increases in definitiveness in proportion to the exhaustiveness achieved in the coverage of Greek forms with -υλλ- and -ολλ-; in this connection, I am happy to acknowledge the assistance of Dr. William Johnson (then Director of Research at the Thesaurus Linguae Graecae; University of California, Irvine), who produced for my inspection an exhaustive list of Greek words with -υλλ- and -ολλ- on the TLG CD-ROM D (Irvine, 1992), using the index program on the Ibycus SPC.

49. For a discussion of this type, see most recently Jasanoff 1994 (there also with justification for the reconstruction with *-h_2-, on which further Hackstein 1995: 316n87).

50. The only other option for μύλλω would be to derive it from a nasal-infix formation (thus e.g. Sihler 1995: 499); but, as seen by Rasmussen (1987: 40), this should have resulted in a *μάλλω, the vocalism of which would then need to be explained as having been influenced by μύλη. Note further that the present analysis of μύλλω accords with the essential *o*-grade character that is well-established for this verb (in IE terms), whether this is best explained (as assumed here) on the basis of Jasanoff's „*hi-conjugation*" theory, or via the competing account based on a dereduplicated intensive *$mé$-$molh_2$-ti (Hackstein 1995: 317, with further references).

51. As for σκάλλω, this could be explained in various ways: it could continue a nasal present (cf. βάλλω); or the form could have arisen secondarily after a set zero grade of the type aor. *σκαλα- < *$sk\underset{\circ}{l}H$-$é/ó$-, cf. σκαλεύω [Hippoc., Aristoph.+] „stir, poke", σκαλίζω [Phryn.] „hoe", σκαλαθύρω [Aristoph.+] „dig, explore" and similar forms; or σκύλλω / σκάλλω might derive from an anit variant *$skel$-, as is suggested by some forms (cf. σκαλμός „thole pin", unless the anit quality results from an early generalization of a Saussure's-Law treatment, cf. *$skol[H]$-mo- in Gmc., IEW 925). I do not find convincing Burrow's attempt to align σκάλλω with Indo-Iranian material meaning basically „scatter" (1987: 59), nor his idea that σκύλλω derives from a root *$sk^{\mu}al$-, with labiovelar (similarly already Hamp 1972: 169f. [arguing for *$sk^{\mu}el$-]).

52. Note the accent of κυλλός: apart from the „pronominal" type (ἄλλος, μέσ(σ)ος, τόσ(σ)ος etc.), which may have an entirely different origin (*$médhi̯$+*o*-, etc.), Gk. *-*i̯ó*- is generally accented (καινός, δισσός etc.), including full grades (πεζός, κοινός); for *kol-$i̯ó$- beside *kol-$nó$-, cf. Hamp 1979 on *$ḱom$-$i̯ó$ (κοινός) beside *$ḱom$-no- (U. kumne, kumnahkle, Osc. *comenei* etc.).

53. As clearly for IIr. *car^{i}*-, cf. EWAia I, 535.

54. On πλήμνη (and Myc. *qe-re-me-ne-u* = /K^{w}lēmneus/) with set zero grade *$k^{\mu}\overset{\circ}{l}h_1$- see recently Plath 1994: 241 (following Waanders).

55. „Von *$k^{\mu}\overset{\circ}{l}nó$- kann sich mit Schwa Secundum neben Labiovelar ein *κυλνός ergeben" (1990: 31).

56. Probably also inscriptional [Θ]υλλοφορον (epithet of Dionysus), if correctly restored, and perhaps the personal name Θυλλινος; see Latte 1966: 334 ad θύλλα.

57. Chantraine s.v. θύλλα: „Doit être une variante phonétique de θαλλ-?".

58. For discussion, see Kimball 1988: 246n19, with further references (to which can be added de Lamberterie 1990: 641-663 [661ff. „Structure de la racine"], García Ramón 1992: 199, Clackson 1994: 118ff.).

59. Cf. Kimball (loc. cit.): „There is [...] no positive evidence for the internal *h_2 in the form of an *o*-grade *$thōl$- < *$dhoh_2l$-".

60. Indeed, *o*-grade would be still more appropriate for [Θ]υλλο- (if it were certain that this first member were based on a thematic formation).

61. Excluded outright, for obvious reasons: dialectal forms of the type Lesb. βολλᾱ (similarly Hom. [Aeol.] ἀολλής) etc., as well as several entirely obscure items attested only once or twice in glosses.

62. E. Masson (1967: 108ff.) points to phonological and semantic problems with this traditional Semitic attribution, concluding that the word is „un élément étranger du vocabulaire grec, sans étymologie claire"; she then offers (1967: 110n1) a tentative comparison with the „groupe [...] des mots κόλλοψ, κολλῦρα, etc." (3.4.1.(ii)), „lui-même d'origine obscure".

63. The „Reduktionsstufe" sometimes set up for Slavic (e.g. *kŭl*-, IEW 612) is apparently based on Sloven. *keljè*, but this is more likely to be a secondary development (thus ESSJa, vyp. 10 [1983] 20). The interpretation of Slav. *$klěji̯$/*$klĭji̯$, which is multiply ambiguous, remains difficult, but the forms certainly point in the direction of an *i*-stem. (I am grateful to David J. Birnbaum for helpful discussion of the Slavic material).

64. Cf. a similar suggestion for θύλλα (3.3.4.[i]), although in that case without analogical restoration of the *i*-stem vocalism.

65. 8x Hom., of which 7x *Il.*; cf. also possessive compounds αιολο- (in the *Iliad*: αιολοθώρηξ, αιολομίτρης, αιολόπωλος).

66. **poleu̯o-* > **poleo-* > (via synizesis) **poli̯o-* > πολλο- (see Meier-Brügger 1992: II. 83, with references).

67. Followed by Sihler (1995: 351f.) and (in the main) de Lamberterie (1990: 632ff.). Among various problematic features of such an explanation (granting the starting-point **poleu̯o-*) are the early *u̯*-loss and the pan-Greek generalization of the synizesis treatment. But for **polh$_1$-u̯-o-*, Saussure's Law should yield **polu̯o-*, not **poleu̯o-*.

68. Phonologically, it may be attractive to consider the development **-oli̯-* > (**-ol'l'-* >) **-ull-* in compensatory terms (as pointed out to me by Bruce Hayes): with the palatalization, /o/ before palatalized /l'/ may have developed into a segment with a final raised portion (cf. comparable assimilatory phenomena with the vowels preceding Russian palatalized consonants); when /l'l'/ depalatalized to /ll/, this preceding segment (with high/back features) could have developed toward /u/ as a way of maintaining contrast otherwise lost with the depalatalization. (The failure of /e/ to undergo a similar raising in sequences **-eli̯-* need not constitute a problem, since front vowels may well have been differently configured in terms of vowel space).

69. Esp. Elean αἰλοτρια (DGE 411, 6th c.), cf. Lejeune 1972: § 156n1.

70. This initial statement will serve for the present, subject to certain tentative adjustments below (hence the intentionally vague status of the following resonant in this formulation); cf. 5.1.

71. On the problem see Blümel 1982: 128, Lillo 1990: 13, both with references.

72. This account has certain consequences for the interpretation of the problematic form Hom. πίσυρες which I hope to take up elsewhere in the context of „schwa secundum" treatments.

73. These forms are traditionally aligned with IEW 1. *tu̯er-* (1100f.), which nevertheless primarily incorporates material meaning „whirl, rush" and the like.

74. In principle, συρμός „sweeping motion, track" could preserve an old *o*-grade action noun **tu̯or-mó-;* but συρμός [Plato+] is just as likely to be deverbative to σῦρω [Hdt.+] (cf. κεύθω „hide" : κευθμός „hiding-place" etc., Chantraine 1933: 134f.).

75. See Chantraine and Frisk s.v. θύρσος (including Nachtr., with references); on the HLuv. form see Marazzi 1990: 396 (with references) and Meriggi 1962: 136, both s.v. *tu̯ars(a)-*.

76. Thus explicitly Forbes (1958: 271f.) and Neumann (1961: 86, 107), arguing in support of the connection made originally by Gelb and Bossert.

77. A marginally possible (but ultimately problematic) candidate might be the weakly-attested **dhu̯er-/*dheu̯r-* „whirl, rage, rush etc." belonging with IEW 4. *dheu-* (266), with forms possibly relatable to intoxication (Gk. ἀθῦρω „play, amuse oneself", Russ. *dur'* „foolishness" etc.).

78. Such a scenario might also help account for the otherwise puzzling phonological difference between θύρσος and τύβαρις „celery pickled in vinegar", the latter likewise thought to be borrowed from HLuv. *tu̯arsa-* (see Neumann 1961: 86f.).

79. Schindler 1972b: 34; most recently, and in great detail, Griepentrog 1995: 117ff. (140f. for Greek).

80. The latter from **θύρσθεν*, a remodeling of **θύρ-θεν* after ἔκτοσ-θεν (Griepentrog 1995: 140f.).

81. Still harder to explain, as Griepentrog notes, is the absence of dual inflection in Greek (as opposed, for example, to Vedic). In this connection, it may be worth considering whether the isolated post-nominal form θύρετρα# (e.g. B 415, referring to the gates of Priam's citadel; post-Hom. θύρετρα specifically = „door casings, door frames", see LSJ s.v.) could be a renewal of a frozen dual **θύρε* < **dhu̯ór-h$_1$*. (See Schwartz 1992: 401 for a different analysis of θύρετρα).

82. E.g. (according to the usual explanation) **ku̯ - p* > *k - p* in καπνός „smoke" and κόλπος „bosom, lap (etc.)" (Lejeune 1972: § 72 n. 3), or (at least in Ionic) early loss of digamma before */o/* and */ō/* (Chantraine 1958: 123ff.).

83. I.e.: „l'élimination de *w* après la dentale aspirée doit être notablement plus ancienne qu'après la sonore" (1976: 52). This entire conception of the treatment of **dhu̯-* (which constitutes, to be sure, a long-standing problem) is rejected by Schwartz (1992) in favor of **dhu̯-* > σ(σ)-; I remain skeptical, however, as to the three etymologies he proposes in support of this treatment.

84. Schrijver's **surh$_2$-ek-* for the Greek form (1991: 128) has nothing to recommend it: while ὕραξ fits well with other Greek -ακ- forms (more on which below), Greek formations in **-ek-* are virtually non-existent.

85. Frisk s.v.: „ohne Etymologie" (similarly Chantraine); none of the proposals cited (and rejected) by Frisk is in the least convincing.

86. In addition to the problematic absence of geminate spellings in Aeolic (cf. Alc. 303Ab.3 Voigt [= Sa. 99b.3 LP/Campbell] ὐλώδη<ν>; despite Leukart, the connection with the tribe name Ὑλλεῖς remains uncertain), it would be difficult in the extreme to align the *l*-formations of *usli* and ὕλη.

87. E.g. H 418 = 420, ψ 50, 111 (Patroclus); ι 234 (the wood taken by Polyphemus, after Odysseus and his men used it in a ritual offering, cf. 231 πῦρ κήαντες ἐθύσαμεν). Cf. also … λάκε δ' ἀμφὶ πυρὶ μεγάλ' ἄσπετος ὕλη# (Hes. *Th.* 694, of the fire resulting from Zeus' thunderbolt).

88. ψ 114 ὑλοτόμους πελέκεας, 123 πάντες δ' ὑλοτόμοι (= agentive „woodcutters"), both with reference to the funeral of Patroclus.

89. I discuss this formation in more detail in a forthcoming paper on „The Etymology of Greek κώμη and Related Problems".

90. Cf. earlier also ϑ' εἰλόπεδον for transmitted ϑειλόπεδον η 123 (despite Hainsworth, in Heubeck et al. 1988: 330, ad loc.).

91. Hsch. βέλα· ἥλιος, καὶ αὐγή, ὑπὸ Λακώνων, γελοδυτία· ἡλιοδυσία, among other material from glosses, beside ἔλη at Aristoph. *V.* 772 [schol. Rav.], where εἴλη appears as a doubtful variant.

92. In addition to the discussions by Beekes (loc. cit.) and in the etymological dictionaries, note also Peters 1980: 86, with the observation that εἰλόπεδον could itself reflect ἐλ- + metrical lengthening.

93. In literary texts, ἔλη otherwise only Eustathius; εἴλη / εἴλη otherwise only Lucian and Alciphron.

94. Thus, for example, εἰλ- / εἰλ- could show a secondary „distraction" (*ἐ-(ϝ)ελ-) back-formed to ἐλ- / ϝελ-, in imitation of pairs like ἔρσαι (ι 222) : ἐέρση (Ψ 598+) „dew" (somewhat similarly Beekes 1969: 76), in which ἔρσαι results from a late Ionic shortening (Peters 1980: 316f.). (It may not in fact be coincidental that εἰλ- / ἐλ- „heat / warmth (of the sun)" is semantically related to ἐέρσ- [i.e. late Ion. εἰρσ-] / ἐρσ- „dew".)

95. Esp. σειρά [Ion. σειρή, Hom.+] „rope, cord, noose" (**tuer-iā*) and derivatives; also σερίδες· σειραί Hsch.

96. Cf. Perpillou 1976: 53n28 for similar doubts as to the semantics.

97. Compare the extremely low percentage of IE terms among Mycenaean vessel names, as calculated by Anderson (1996), who registers a double question mark for the IE status of Hom. σορός. On non-IE vessel terms in Greek (from a historico-cultural perspective) see recently Drews 1988: 178.

98. Beside σύριχος / συρίσκος, also ὑριχός, ὑρισσός / σύρισσος, ὕρισχος, etc. (see LSJ s.v. ὑριχός).

99. E.g. Dunkel (1982b: 58n18), declaring (v) to involve a „match" (scil. between ὕμνος and Ved. *sumná-*) that is „irresistable" [sic]; Peters is notably non-committal.

100. In some cases the problems are mainly semantic ([i] and [ii]) or phonological (as in [iii]), and (vi) is merely a counsel of despair (Chantraine: „une échappatoire peu plausible"). The others involve more complex considerations, including the following: (iv) founders on the questionable status of ὑδέω [Callim.+], see Peters 1980: 65ff.; (v) depends, firstly, on a treatment of **h₁su-* that is far from certain (see Weiss 1993: 169ff., Leukart 1994: 99n182), and secondly, on the possibility of taking *sumná-* to mean „Huldigungslied" or „Gebet" at RV 3.14.4b (see KEWA s.v. *sumnám,* with refs., and EWAiai II, s.v.), which is at best unnecessary (see Geldner ad loc. on the comparison between *sumnám arc-* [3.14.4b] and *sunatím arc-* [4.4.8a], with Sāyaṇa's interpretation of the latter); (vii) requires a generalized form of Cowgill's Law (similar to that proposed by Dunkel for σύν), for which there is no evidence (cf. 2.7.); moreover, the suffix formation seems to require an unattractive vacuous thematization of an *n*-stem (cf. Ved. *sāman-* „song"; an appeal, alternatively, to other Greek forms with apparent suffix **-no-* [στάμνος „jar", θάμνος „bush" etc., thus explicitly Frisk] is of little value, as these words are for the most part obscure: Chantraine 1933: 191ff., 215); nor, finally, does Benveniste's analysis of οἶμη „song" (allegedly < **sh₂om-i̯o-*)

provide much support: as pointed out by Beekes (see Lindeman 1987: 49n30), **sh₂om-i̯o-* should yield -οιν-, not -οιμ- (cf. κοινός < **kom-i̯o-*).

101. Thus typically LSJ („hymn, ode, in praise of gods or heroes") and Chantraine („[…] notamment en l'honneur des dieux […] semble avoir eu d'abord une coloration religieuse").

102. Not surprisingly, in view of the ordinary meaning of ὕμνος in later Greek, commentators have been puzzled by this usage; note e.g. Stanford (1959: 343 ad loc.), observing „ἀοιδῆς ὕμνον: „hymn of a ballad", a curiously elaborate phrase", and the remarkable assertion of Hainsworth (in Heubeck et al. 1988: 375 ad loc.), that „In the strict sense ὕμνος celebrates a god, and […] it is not a proper term for heroic narrative poetry, such as is appropriate to the banquet"; cf. also Nauck's conjecture οἶμον, based on h. Merc. 451 ἀγλαὸς οἶμος ἀοιδῆς# (with v.l. ὕμνος for οἶμος).

103. „I won a contest in song" (transl. Calame 1995: 210).

104. See now also Ford 1992: 24 („early uses of *humnos* and the verb *humnein* can refer to any kind of song"), and 1992: 112n29 for speculations as to semantic distinctions between ὕμνος and other terms for song, esp. ἀοιδή.

105. On the metathesis (with special reference to Ἀγαμέμνων), see Hamp 1971b: 21ff. For the specialization of **su̯enH-* in the direction of song and instrumental music, cf. OIr. *sennid* „plays (an instrument)", OE *swinsian* „sing, play music" and Eng. *swan* („singer"), as well as episodic usages like RV 8.69.9b *godhā́ pári saniṣvaṇat* (referring to the playing of a stringed instrument).

106. ἱπποσύνη „horsemanship", γηθοσύνη „joy" and another dozen or so such forms in Homer (surveys in Risch 1974: 150, Powell 1988: 104ff.).

107. RV *mahitvaná-* „might, lordship", *vasutvaná-* „abundance", *sakhitvaná-* „friendship" and another five forms in the Rig-Veda (survey in Hopkins 1896: 37).

108. Recently e.g. Devine and Stephens 1994: 104, with reference to -σύνη and *-tvaná-* as one of the prime examples of Bartoli's law of accent retraction (cf. θυγάτηρ : Ved. *duhitā́* „daughter", among other such pairs).

109. 1935: 115f. (substantially following Brugmann).

110. Wyss (similarly already Porzig and others, see Risch, loc. cit.) begins from the adj. Hom. γηθόσυνος „glad, rejoicing (in)", which he plausibly takes to be formed on the model of θάρσυνος „encouraged (by)" and πίσυνος „trusting (in)", and builds his account around the parallel existence of γηθόσυνος and γηθοσύνη „joy", variae lectiones of the sort γηθοσύνη (abstr.) for γηθοσύνη (adj.) (e.g. λ 540) and vice versa (e.g. N 29; see Janko 1992: 46 ad loc.), and a conjectured phrase like *γηθοσύνη φρήν, which in turn leads to the Homeric words in -φροσύνη (seen as the basis for the rest of the -σύνη vocabulary). Against this it may suffice to note that the -φροσύνη material (mainly *Od.*, and almost entirely without evidence of traditional phraseology) seems to belong to the latest stratum of -σύνη forms, as opposed, for example, to isolated forms denoting skilled activities (mainly *Il.*, and highly formulaic, e.g. ἱπποσύνη, μαντοσύνη „prophetic skill"), which are evidently old (as Powell himself admits, 1988: 123).

111. Note the skepticism voiced by Lejeune on this point (1972: § 52a).

112. In itself not a new idea, cf. Fay 1911.

113. Cf. also RV 8.25.18c, where *...mahitvá#* closes what appears to be an 11-syllable pāda of an Uṣṇih stanze, whence various suggestions for restoration or emendation: in addition to Arnold's [...] *mahitvanā* (1905: 312, ad loc.; cf. [...] *mahitvanā#* 8.24.13c, also Uṣṇih), note Lanman's distracted reading „*mahitvāā*" (1878: 335).

114. A detailed treatment of the historical analysis of this sequence (e.g. **-ṯu̯-onā*, cf. the suffix of ἡδονή „pleasure" etc.?) cannot be undertaken here.

115. The idea of appealing to some version of Cowgill's Law to explain Gk. -σύνη has been considered by Sihler (in draft versions of Sihler 1995; personal communication), and seems also to have been envisaged by Dunkel, to judge from a laconic note (1982b: 59n25) that does not, in any case, clarify the details of his conception.

116. Ruijgh 1967: 86. Cf. also Hom. „-οφι", on which recently Clackson 1994: 68ff.

117. One would especially like to understand why the Kunstsprache ended up choosing (or generating) a formant that could not in fact be used as such, but which required „metrical shortening" by degemination in order to be used in dactylic verse.

118. This formulation is conceived of in Proto-Greek terms, as laryngeal clusters **Hu̯-* (with the laryngeal already vocalized by the time of Proto-Greek) are evidently not affected (thus e.g. **h₂u̯olk-* > **au̯olk-* > [acc.] ὦλκα „furrow").

119. On Middle Indic *supina / soppa* as resulting from an inner-Indic samprasāraṇa see Hinüber 1986: 82.

120. See in general Schindler 1966: 74 n. 26, Hamp 1970b: 143, Watkins 1972: 558 n. 13, Nussbaum 1986: 14, Barton 1993: 554 with n. 26.

121. E.g. Ernout - Meillet s.v. *somnus*: zero grade in Gk. and Slavic secondary after ἐνύπνιον [B 56+], *sŭnĭje* „dream".

122. This is not to say that all of the details of Barton's account are equally convincing, but the subject cannot be pursued further here.

123. Here note also the Hitt. syncope type *CwV* > *Cu*, in some cases „associated with extra word-length or a shift in accent caused by the addition of further suffixes" (Melchert 1984: 53).

124. Among zero-grade material such as that assembled by Schwyzer 1939: 518f., the best candidate appears to be κύαρ „eye (of a needle), hole" [Hippoc.+], especially if it is to be equated (as **ḱuH-r̥*) with Toch. B *kor* „throat" (thus Adams 1991: 5n9), cf. also RV *śū́na-* „emptiness" beside YAv. *sūra-* „hole", possibly pointing to an old heteroclisy **ḱuH-r-/*ḱuH-n-* (cf. EWAia II, 650). Whether Arm. *sor* „hole" belongs with this *r*-stem is uncertain (in view of non-original *r*-formations like *nor* „new"); otherwise, the general absence of full grades (except in some related thematic forms without **r / *n*, of the type Lat. *cavus* „hollow" < **ḱou̯H-o-*) is surprising, and may be systematic. If these forms belong (as is usually thought) with **ḱeu̯h₁-* „swell" (Gk. κυέω etc.), the unexpected zero grade in κῦμα „wave, swell" is perhaps noteworthy,

and may serve to weaken still further the value of κύαρ as a genuine exemplar of a type of archaic zero-grade *r/n*-stem.

125. It is evident, of course, that the vocalism of the isolated ὕπαρ cannot be explained analogically after that of ὕπνος.

126. See now Melchert 1994: 49f. for discussion of Eichner's *o*-grade interpretations of *uttar* (**h₂u̯ódh-r̥*) and *watar(-nahh-)* (**h₂u̯ód-r̥*).

127. For the connection see now de Lamberterie 1990: 173, EWAia I, 639 (s.v. *TAV*[1]), Clackson 1994: 177.

128. Note also σόος (for σῶς), with secondary „diectasis" influenced by σάος. The interpretation of σάος itself remains difficult, but the form does not in any way undermine the above claims regarding the treatment of /Cu̯o/. At present, assuming the connection with Ved. *távīti* etc., I incline toward the „reverse Lindeman" analysis as per Peters 1980: 290n243, i.e. **tu̯áu̯os* < **t(u)u̯áu̯os* < **tuh₂éu̯os*. (The remade zero grade **twă-* posited by de Lamberterie [1990: 177] depends on a „thème II" **tw-eh₂-*, which again raises the problem of Schwebeablaut.)

129. γυνή in any case does not show **Kᵘe* > **Kᵘo*, for which only κύκλος is a possibility.

130. Thus Sihler's concern as to why κολοφών „summit" does not end up as †κολυφών (1995: 43; explained via an original **κολα-* < **koln̥-*) or as to why **polh₂-meh₂* ends up as παλάμη and not †πυλάμη (1995: 111) involves non-problems to a greater extent even than his discussions imply (entirely apart from the question of whether these are the correct preforms for these words).

131. In accordance with 4.6.(i), it is possible that this process should be restricted to accented syllables.

132. This paper was originally presented at the 14th East Coast Indo-European Conference (Cambridge, Massachusetts; July 1995), which was devoted to the memory of Jochem Schindler. I am grateful to the participants for their helpful comments, as well as to audiences at subsequent presentations before members of the UCLA Program in Indo-European Studies and the UCLA Phonology Seminar. This work was supported in part by a UCLA Senate Research Grant.

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Questions of Syntax and Meter in Tocharian

Calvert WATKINS

The impetus toward this paper, which truly and literally represents work in progress, came from a handout I made for a Tocharian class a few years ago, with the title „Materials for the study of Tocharian poetics". I simply took a familiar verse quatrain or strophe of 14-syllable lines in the Puṇyavantajātaka in Tocharian A, and arranged them as 8 hemistichs of 7 syllables each:

1. **tsrasiśśī** māk nispalntu
tsrasiśśī māk śkaṃ sñaṣṣeñ
nämseñc yäsluṣ **tsrasisac**
kumseñc yärkant **tsrasisac** |
tsrasiñ waste wrasaśśi
tsrasiśśī mā praski naṣ
tämyo kāsu **tsraṣṣune**
p$_u$kaṃ pruccamo ñi pälskam ||

The strong have great riches; **the strong** have also many relatives.
Enemies bow down to **the strong**; honors come to **the strong**.
The strong (are) the protection of creatures; **the strong** have no fear.
Therefore **strength** (is) good; best of all in my opinion.

The resulting pattern shows on the one hand that the clear and obvious classical rhetorical figure of anaphora (that in, e.g., Lincoln's „of the people, by the people, for the people") is associated with adjacency to metrical boundary (verse line or hemistich; the repeating words are marked in boldface). On the other hand it shows that the finite verb as well (only three here to be sure) is found in the same two positions: immediately preceding metrical boundary, i.e. colon-final, as here with *naṣ*, or immediately following metrical boundary, i.e. colon-initial, as here in the contrastively focused *nämseñc, kumseñc*. We recognize immediately characteristic features of the Indo-European mapping of syntax onto metrical units: the conventional equation of verse-line and sentence, and the familiar unmarked sentence-final and marked sentence-initial position of the verb reconstructed for the parent language since Delbrück.

Not only in form but also in theme this strophe can be compared closely with other and older Indo-European poetical traditions: compare the verses from the Rigveda (6.75.2) and Archilochus (2 West IEG) which I cited already in Watkins 1995: 21-22, again with focused anaphoras in boldface:

2. ἐν **δορὶ** μέν μοι μᾶζα μεμαγμένη· ἐν **δορὶ** δ' οἶνος

COMPOSITIONES INDOGERMANICAE IN MEMORIAM JOCHEM SCHINDLER

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